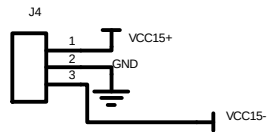


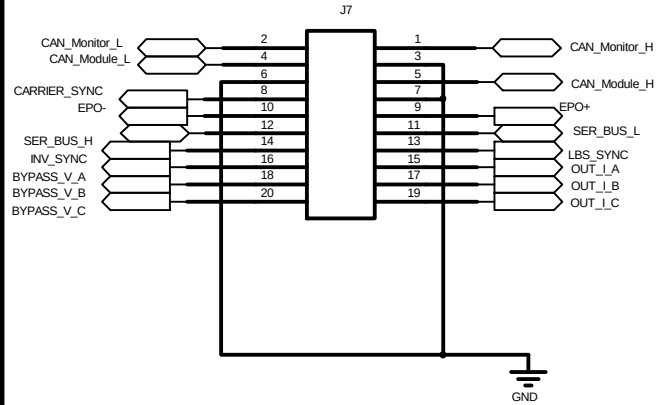
	1	2	3	4	5	6	7	8	
F									F
E	SA500 DSP Detect Board								E
D	Ver 2011.04.28								D
C									C
B									B
A									A
	1	2	3	4	5	6	7	8	

0	NEW	xxx			2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION			Date
PART NO. xxx	SHEET. 01 / 36	TITLE. XXX Circuit Diagram		DRAWING NO. DMBMX.XXX.XXXDL	
Designer	xxx			INNOVPOWER	
Reviewed	xxx	PRODUCT. xxx			
Process Reviewed	xxx				
Standardized	xxx				
Approved	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER			

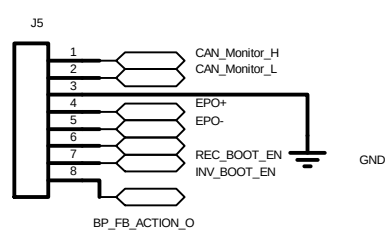
Power in for desktop software debug



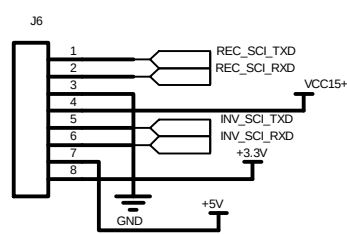
Parallel interface



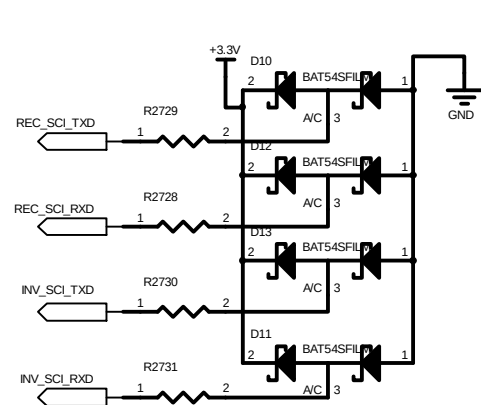
Monitor Board Interface



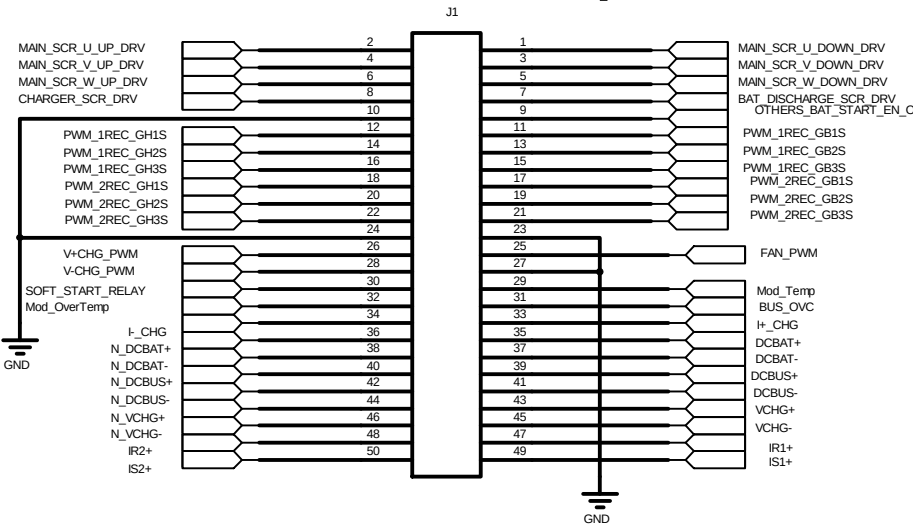
SoftwareDebug Board Interface



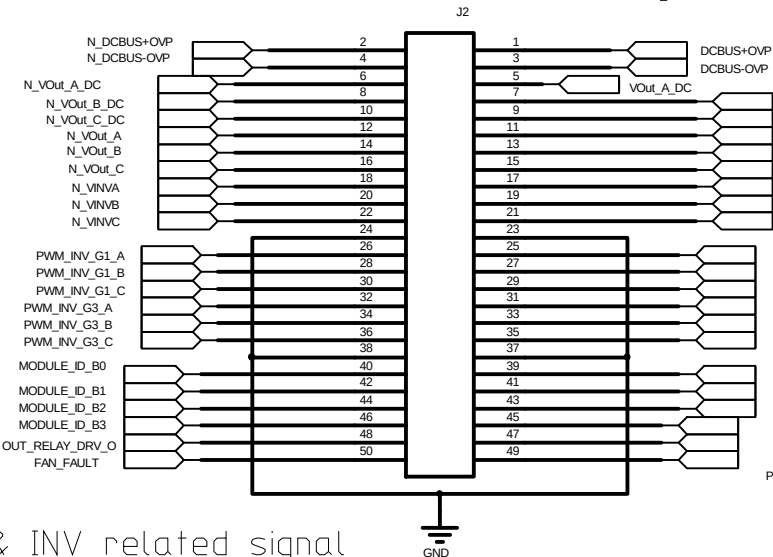
POWER , REF GEN



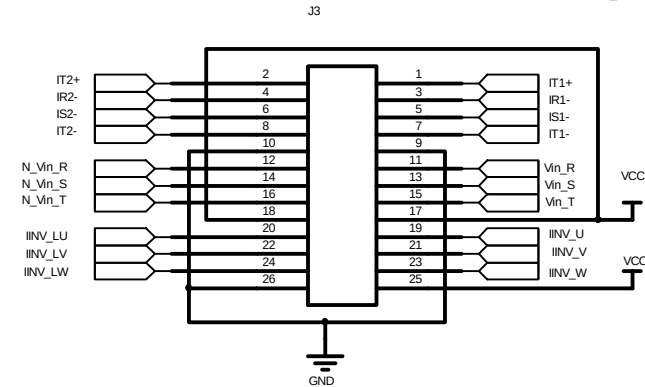
REC related signal



INV related signal

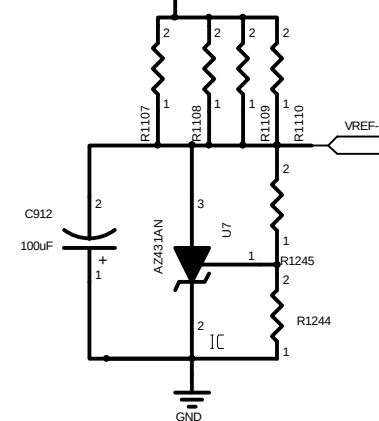


Other REC & INV related signal

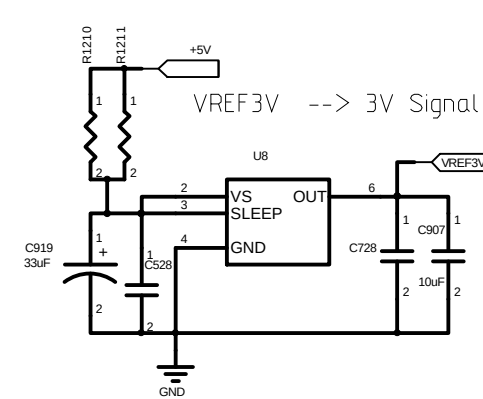


Voltage reference

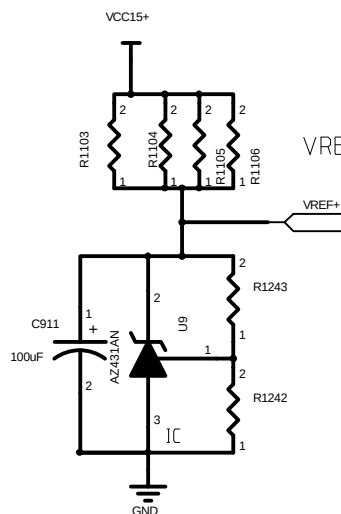
VREF- --> -5V Signal



VREF3V --> 3V Signal

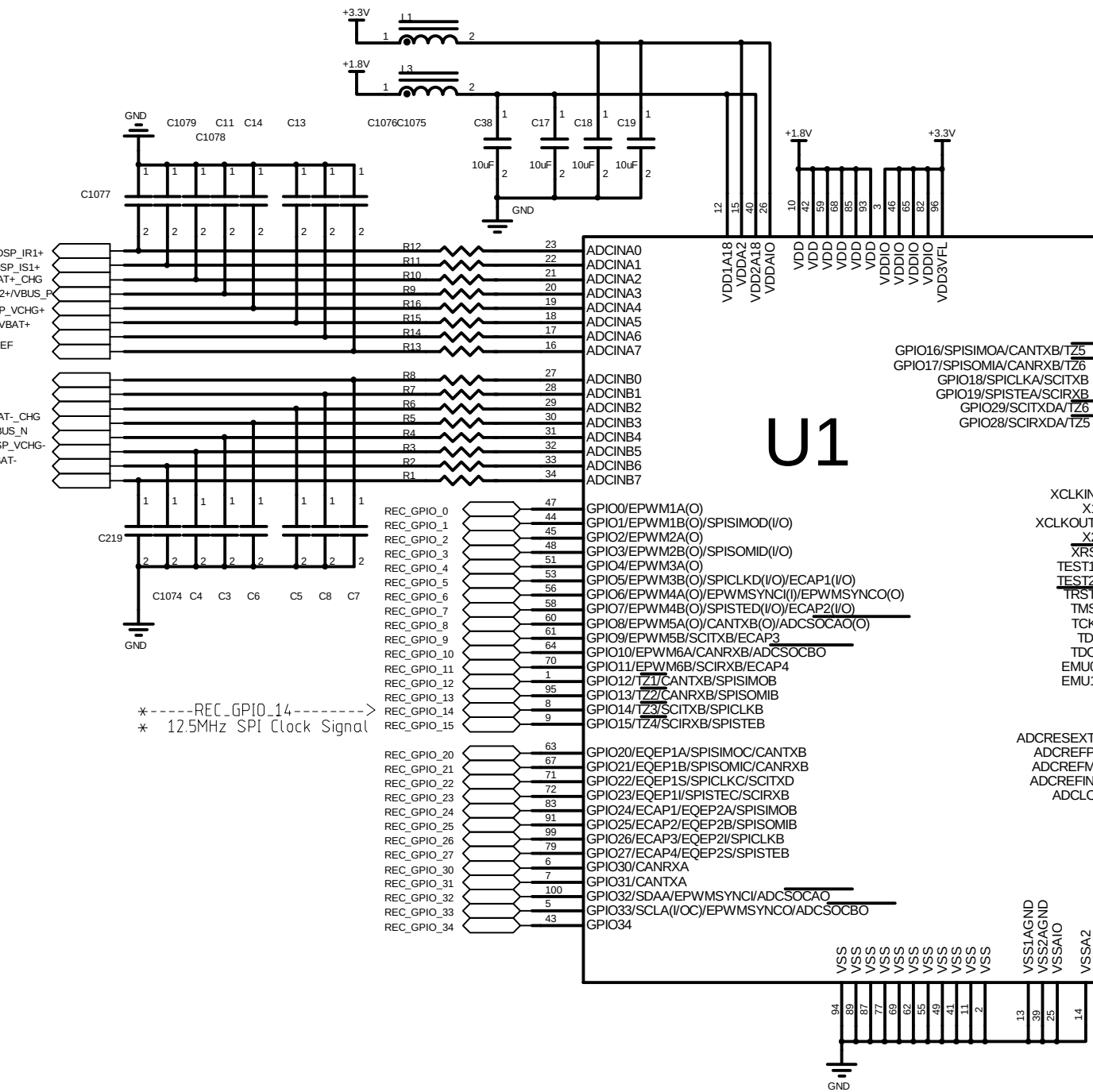


VREF+ --> +5V Signal

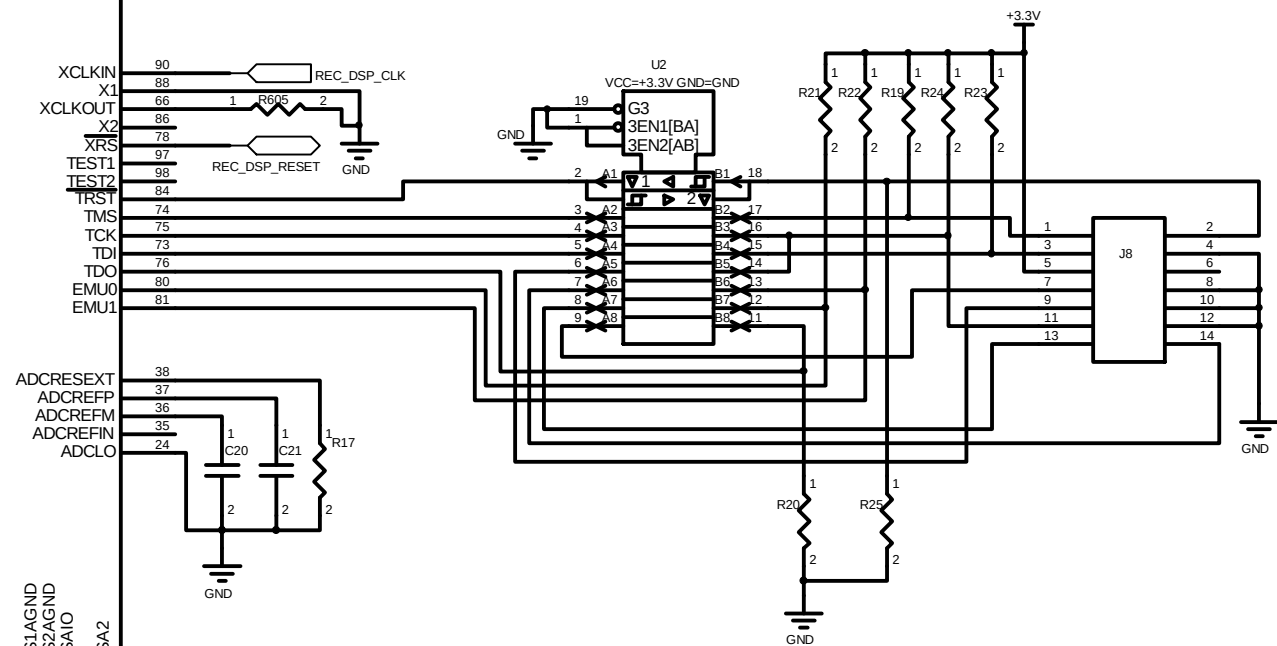
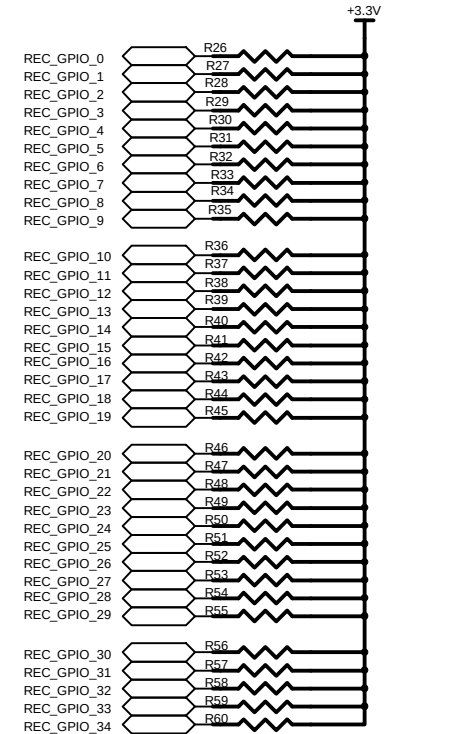
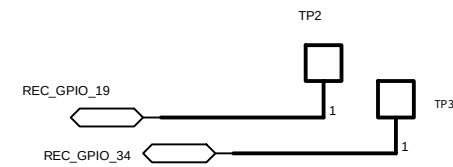


0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	02 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	
Process Reviewed	xxx		
Standardized	xxx		
Approved	xxx		

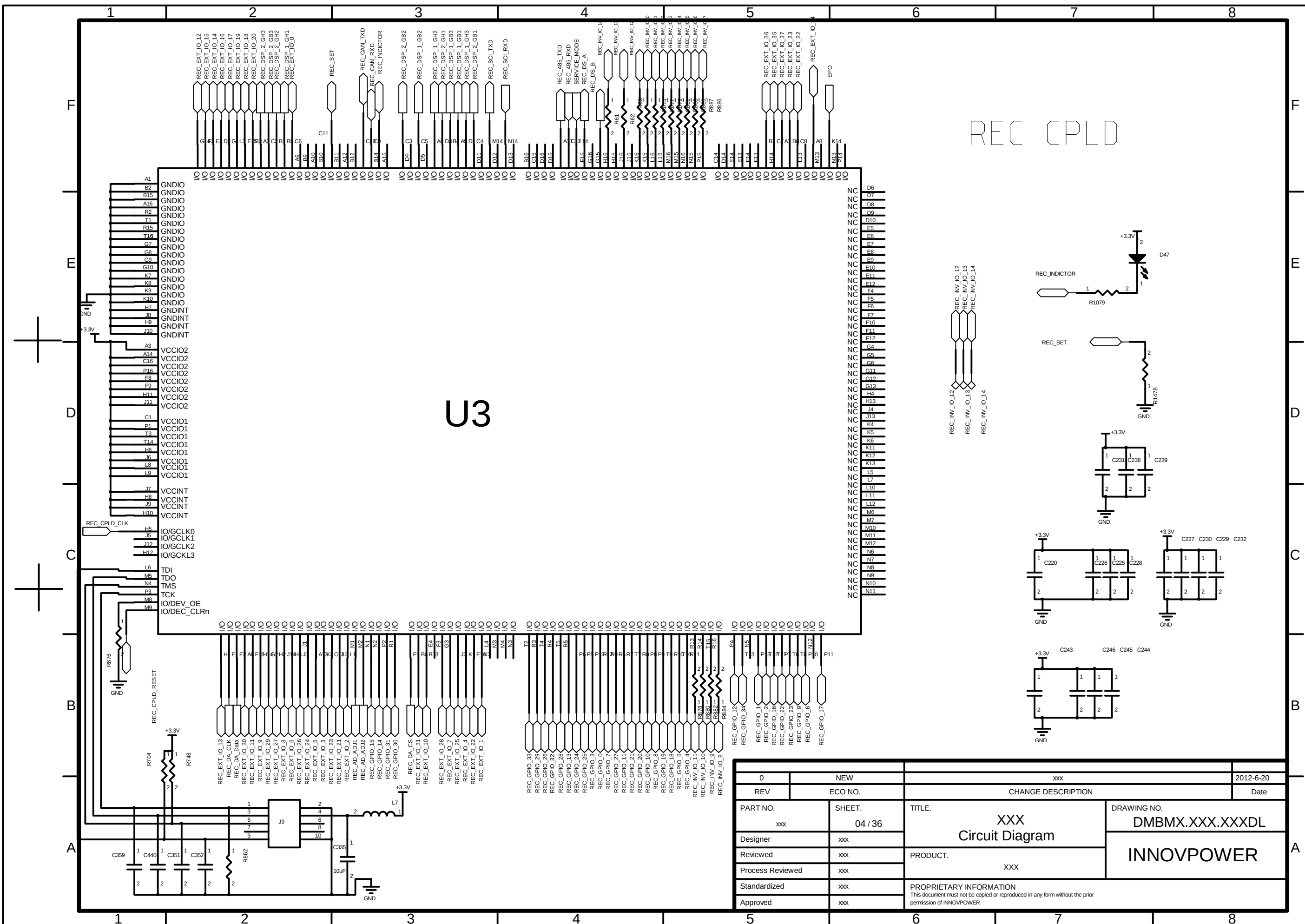
REC DSP

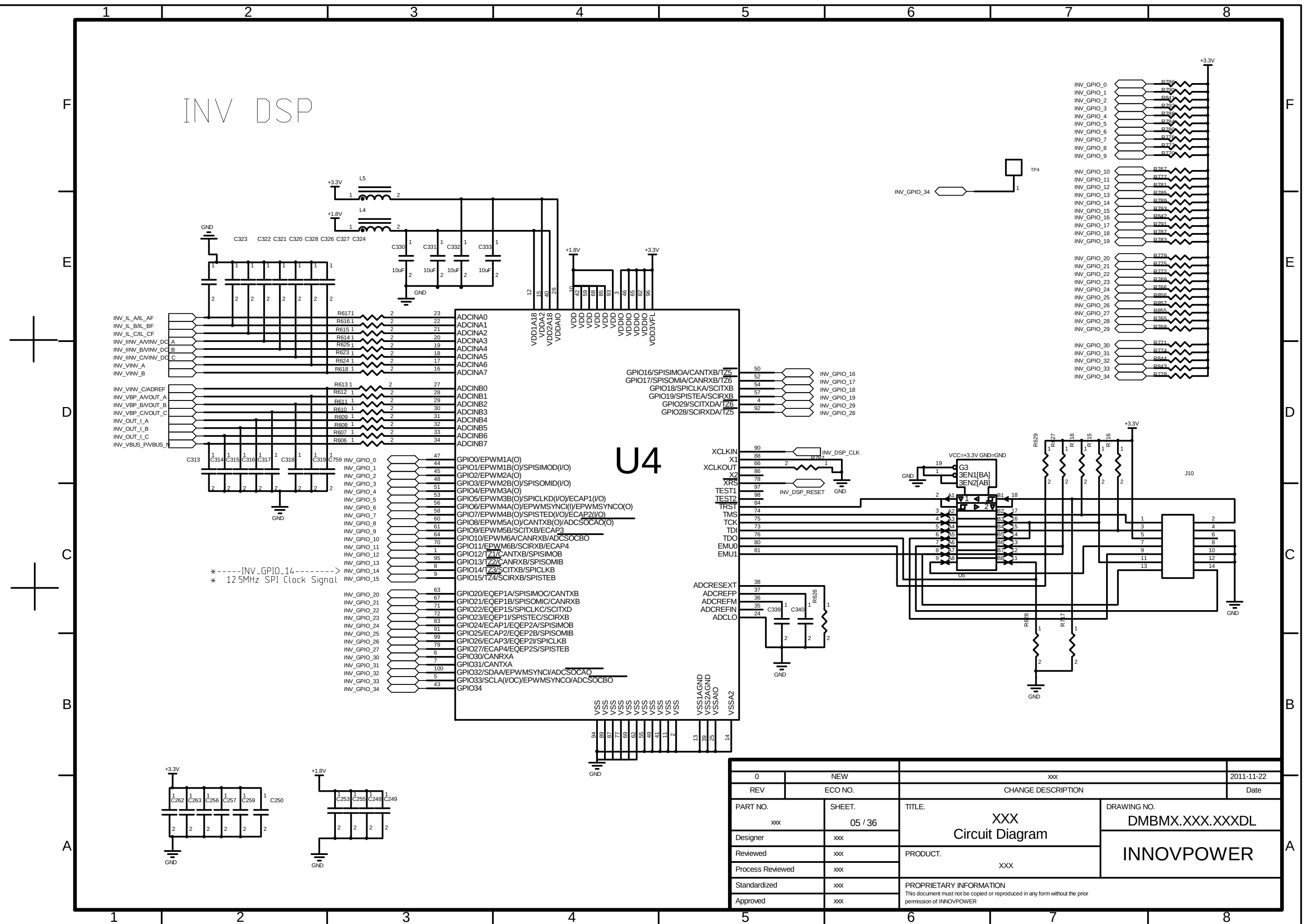


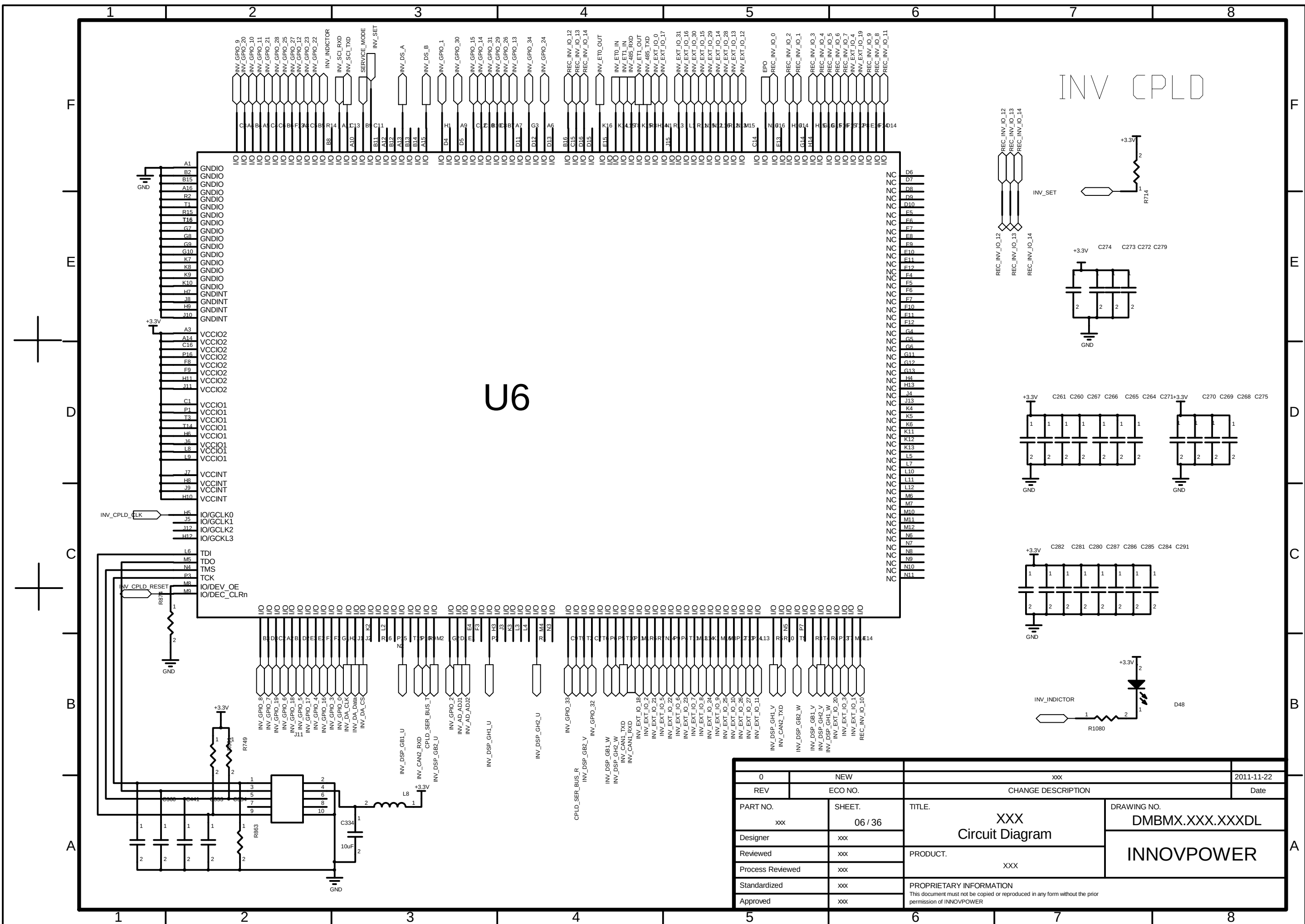
*-----REC_GPIO_14----->
* 12.5MHz SPI Clock Signal

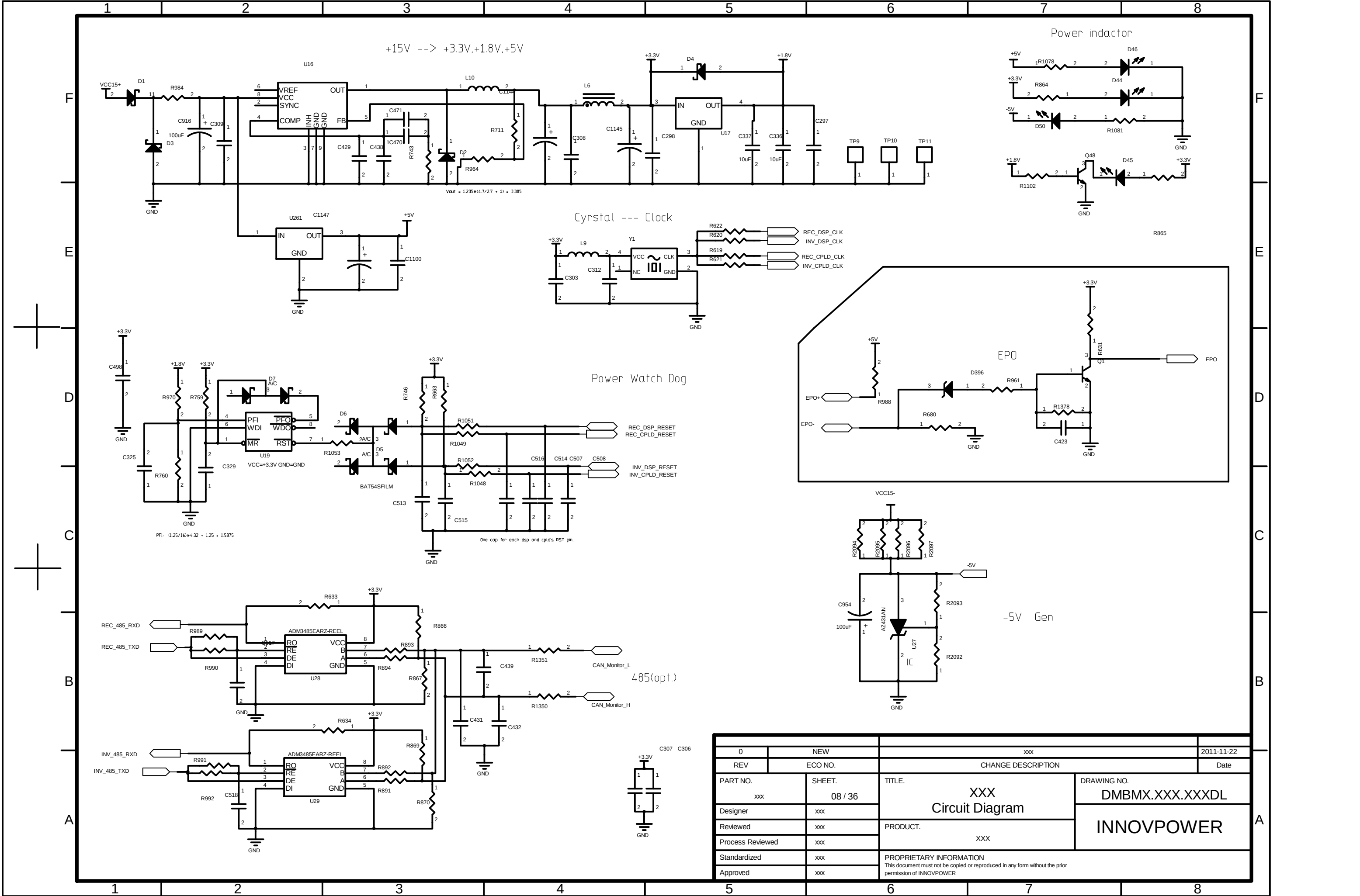


0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	03 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	

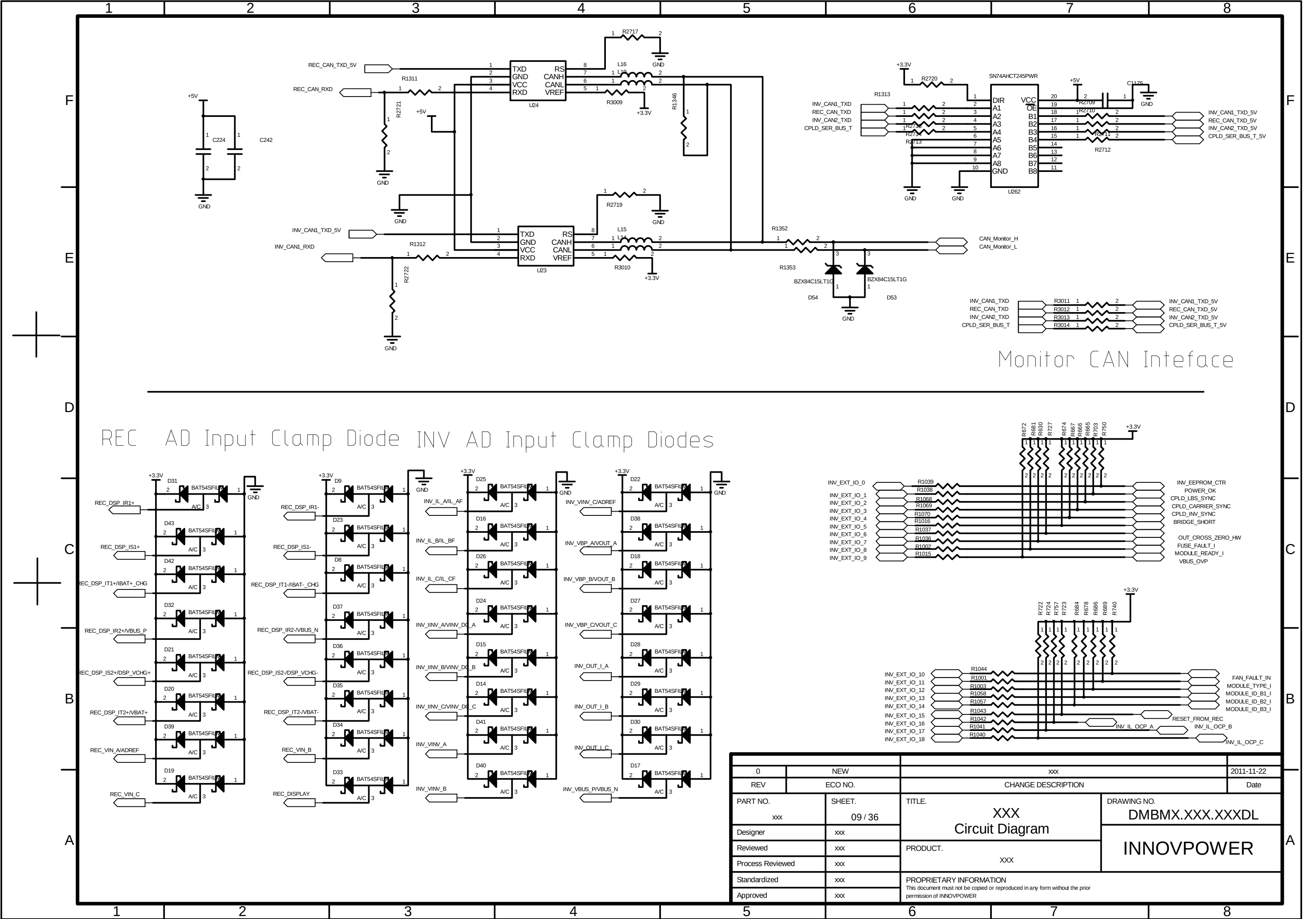


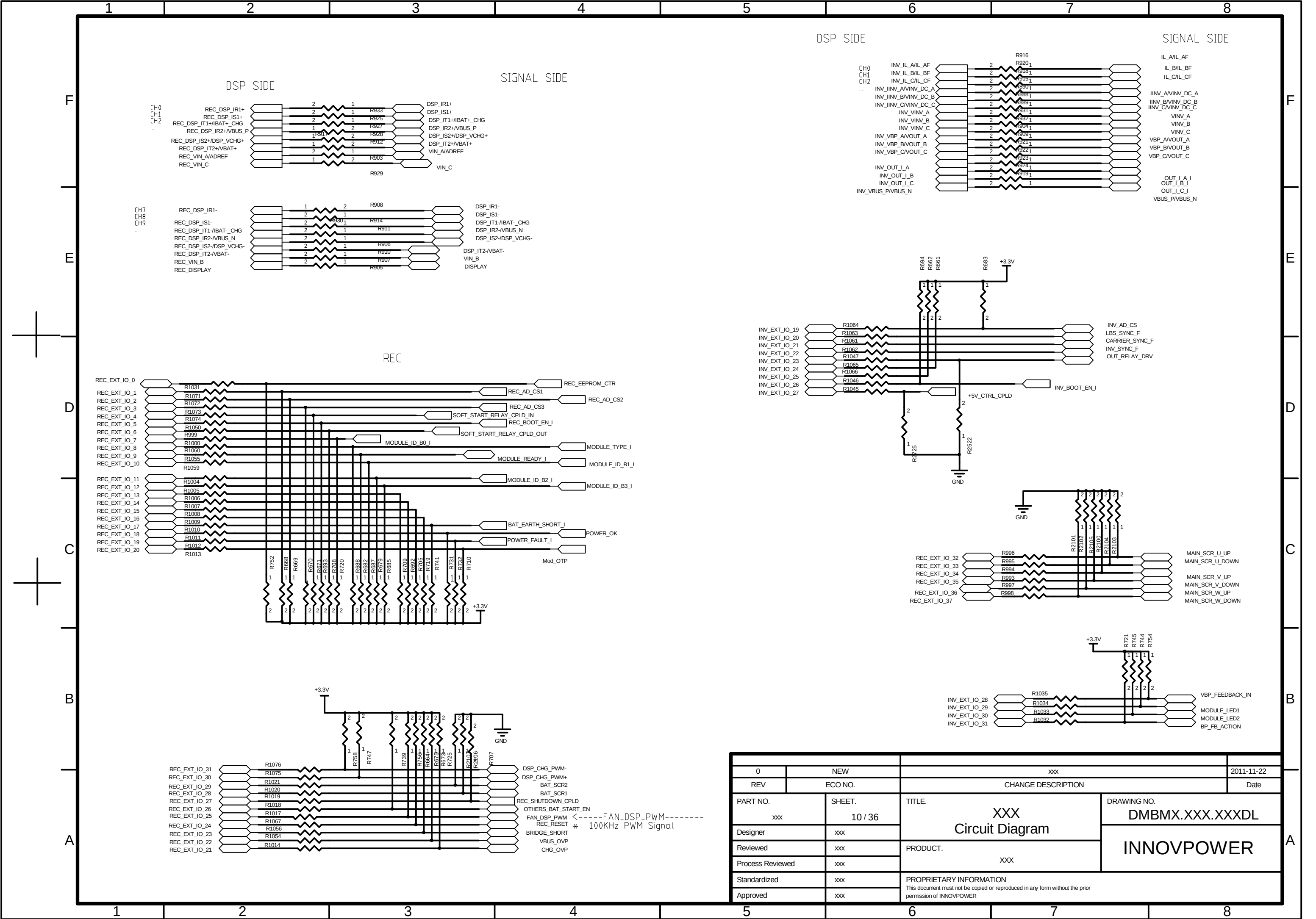




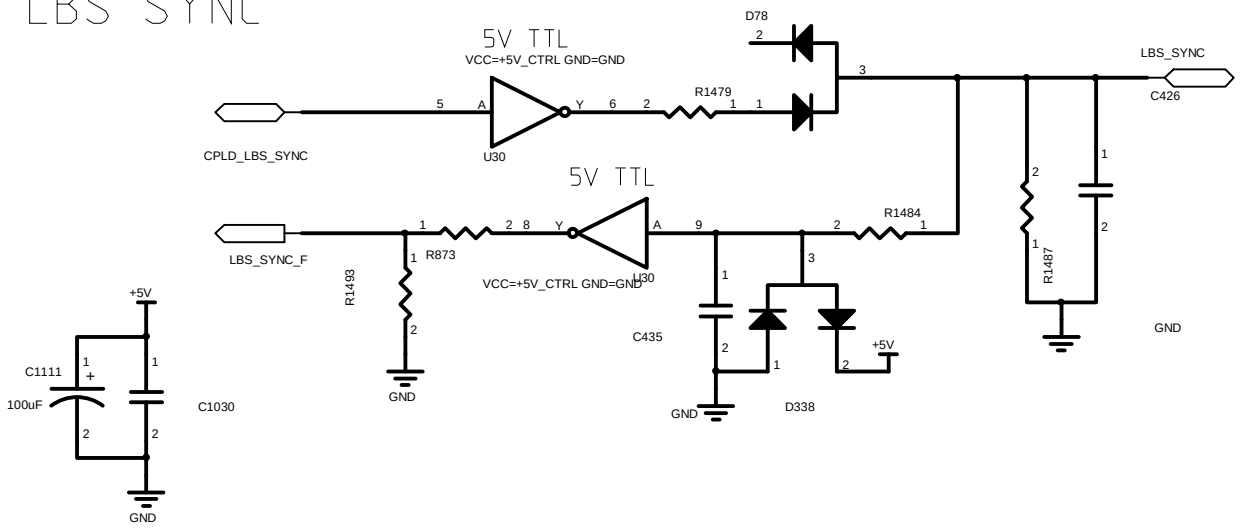


0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	08 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	

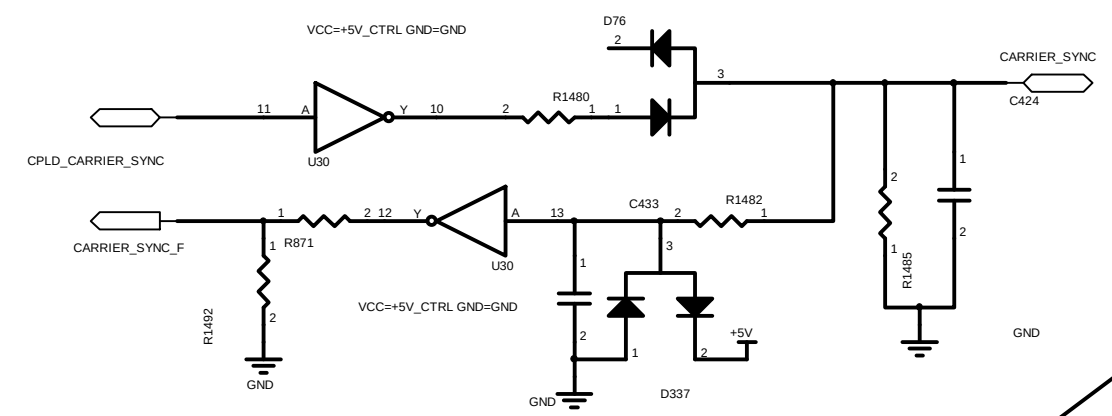




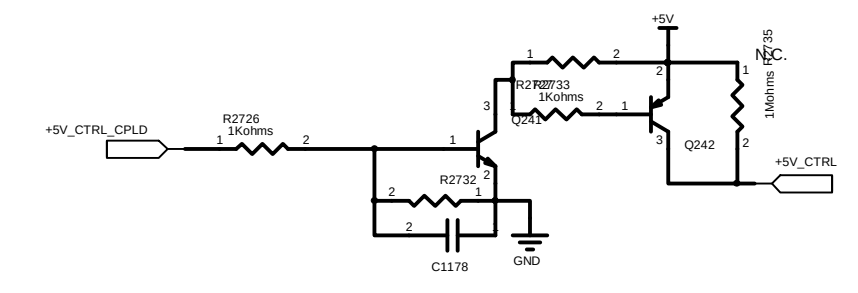
LBS SYNC



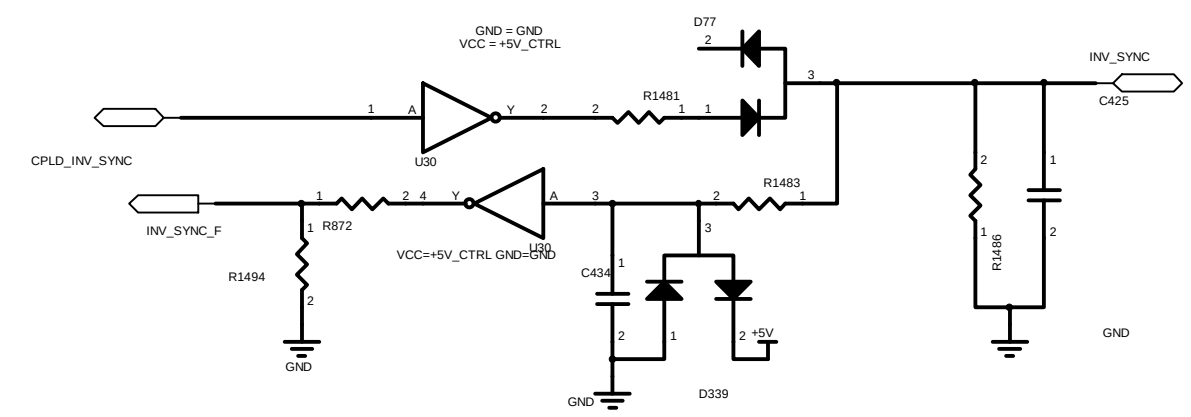
CARRIER_SYNC



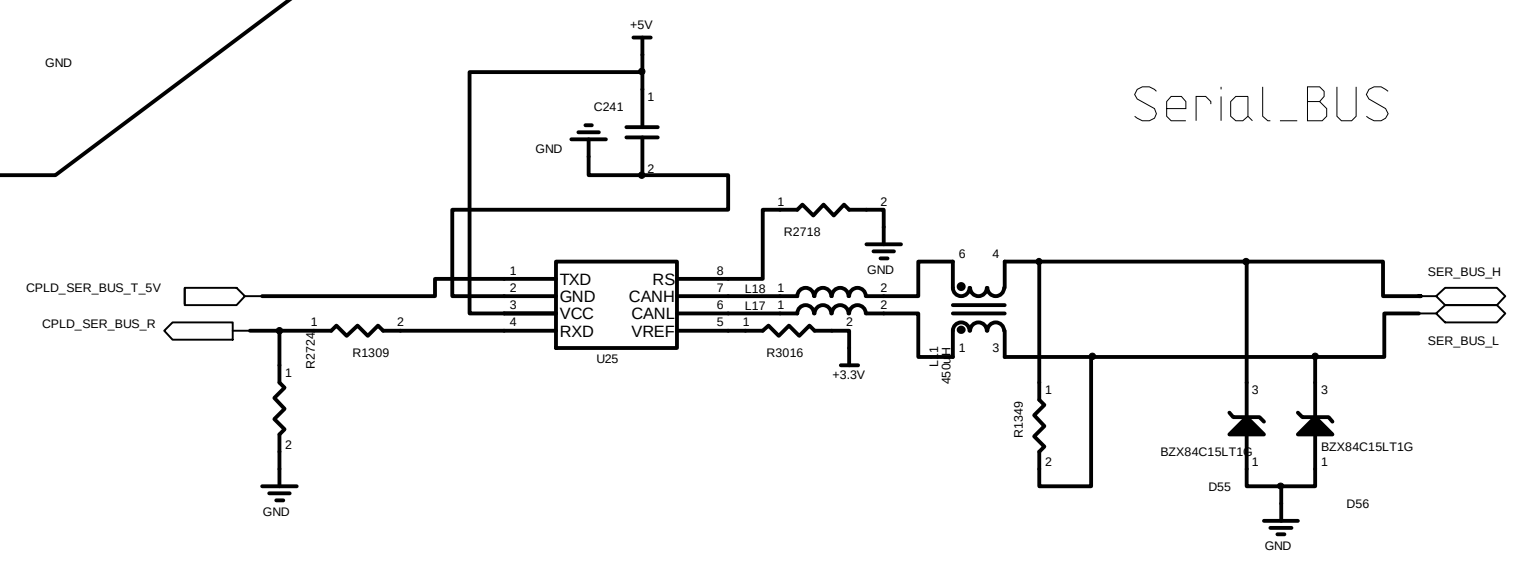
3 Parallel Signal Conditioning



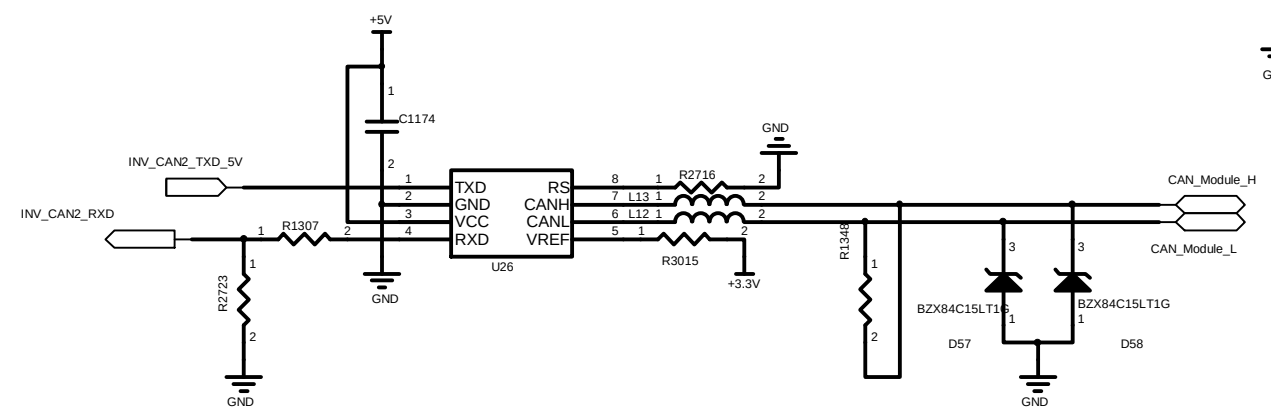
INV SYNC



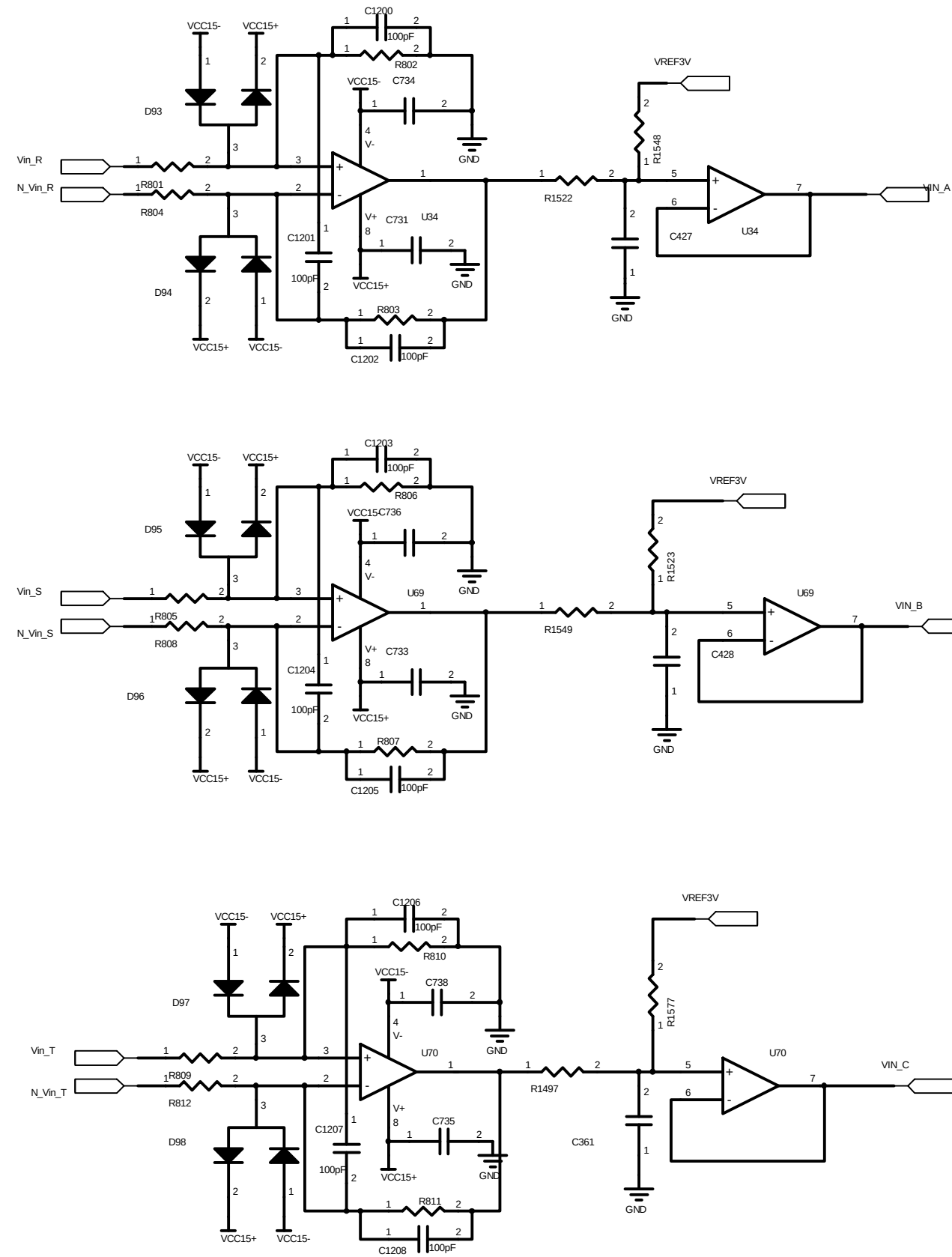
Serial_BUS



Module CAN_Module

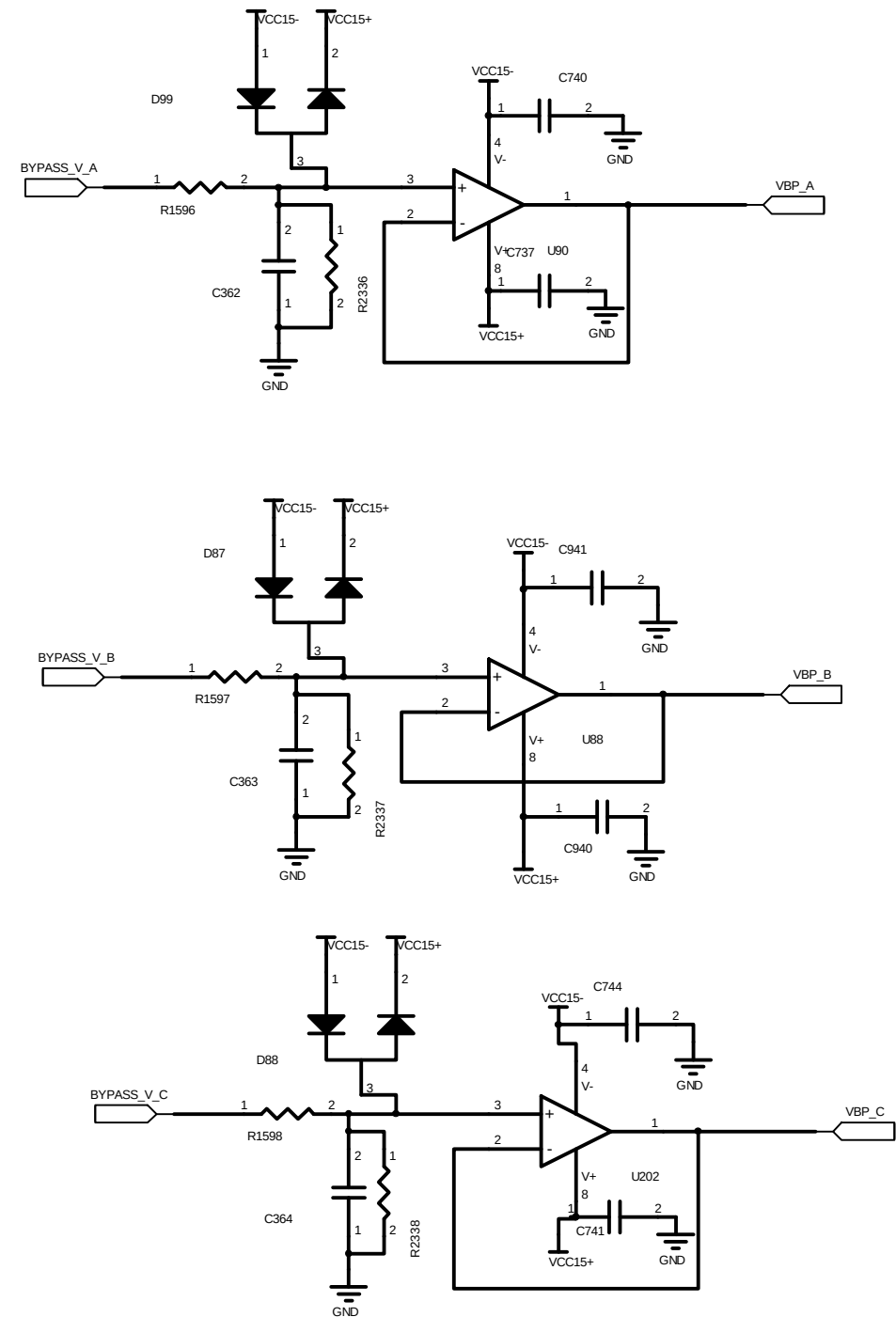


0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	11 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	XXX	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	



Main input voltage detect

-545V to 545V



Bypass Voltage Detect

0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO. xxx	SHEET. 12 / 36	TITLE. XXX Circuit Diagram	DRAWING NO. DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT. xxx	INNOVPOWER
Reviewed	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	
Process Reviewed	xxx		
Standardized	xxx		
Approved	xxx		

1 2 3 4 5 6 7 8

F

F

E

E

D

D

C

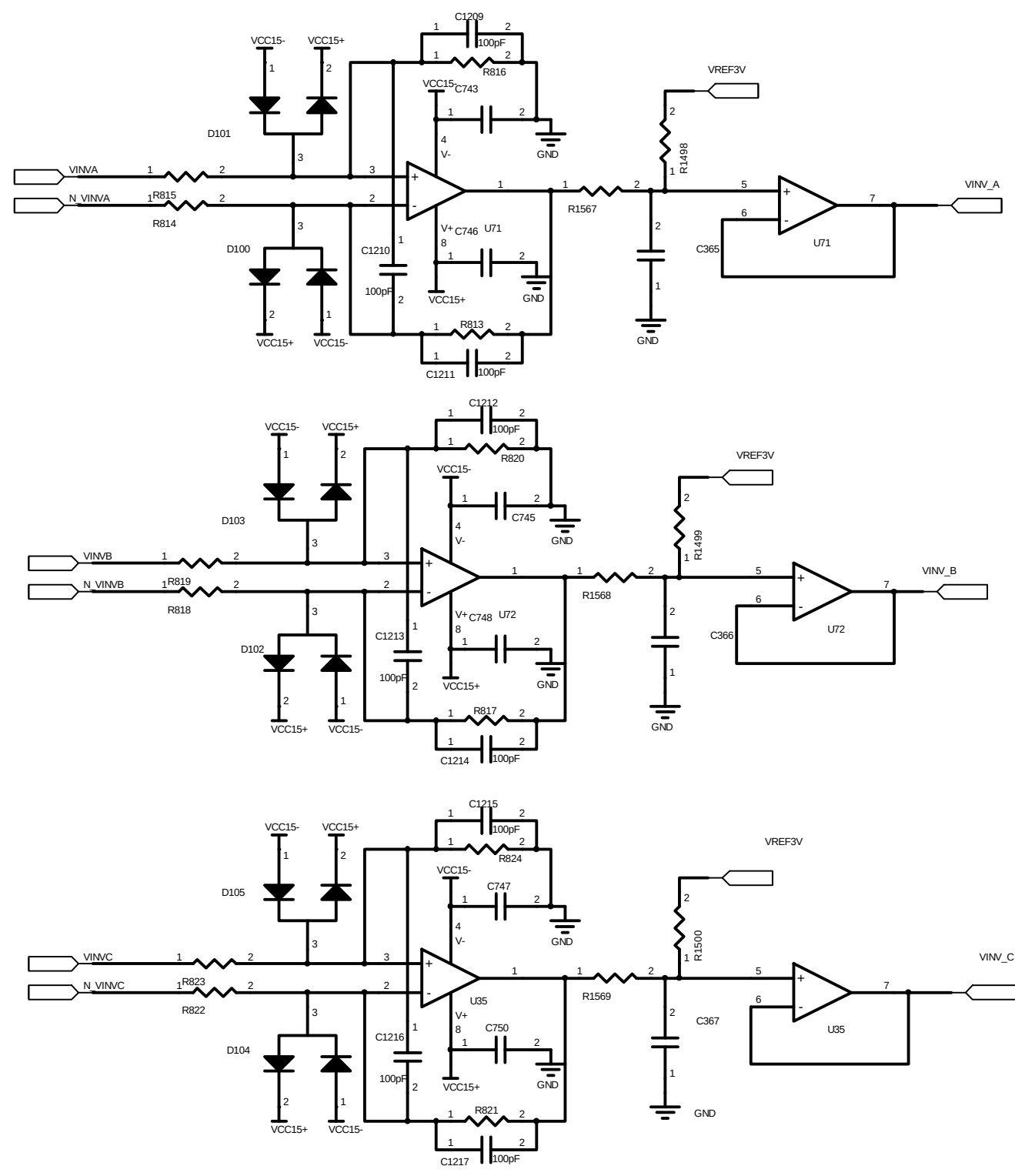
C

B

B

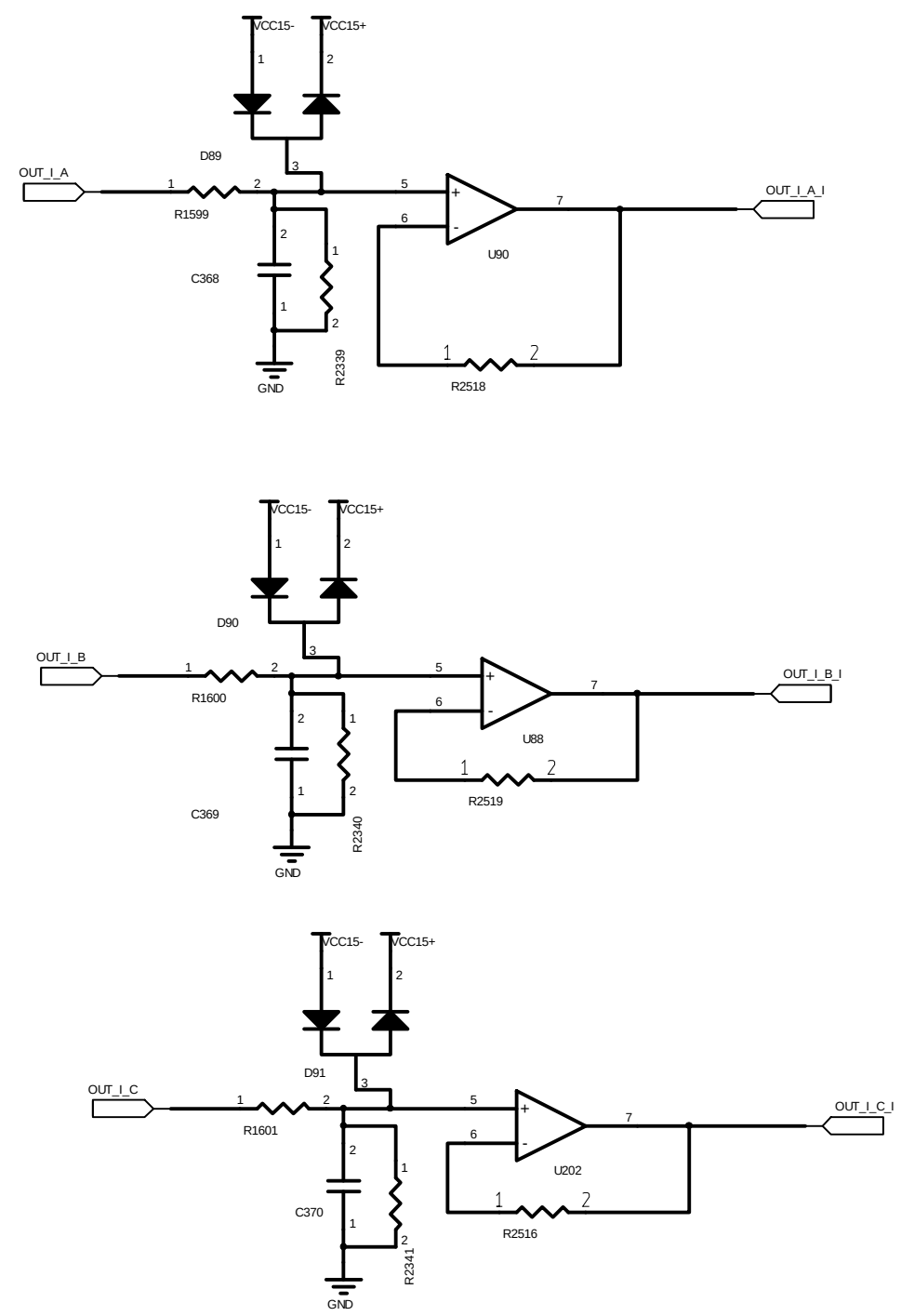
A

A



INV Voltage Detect

-545V to 545V



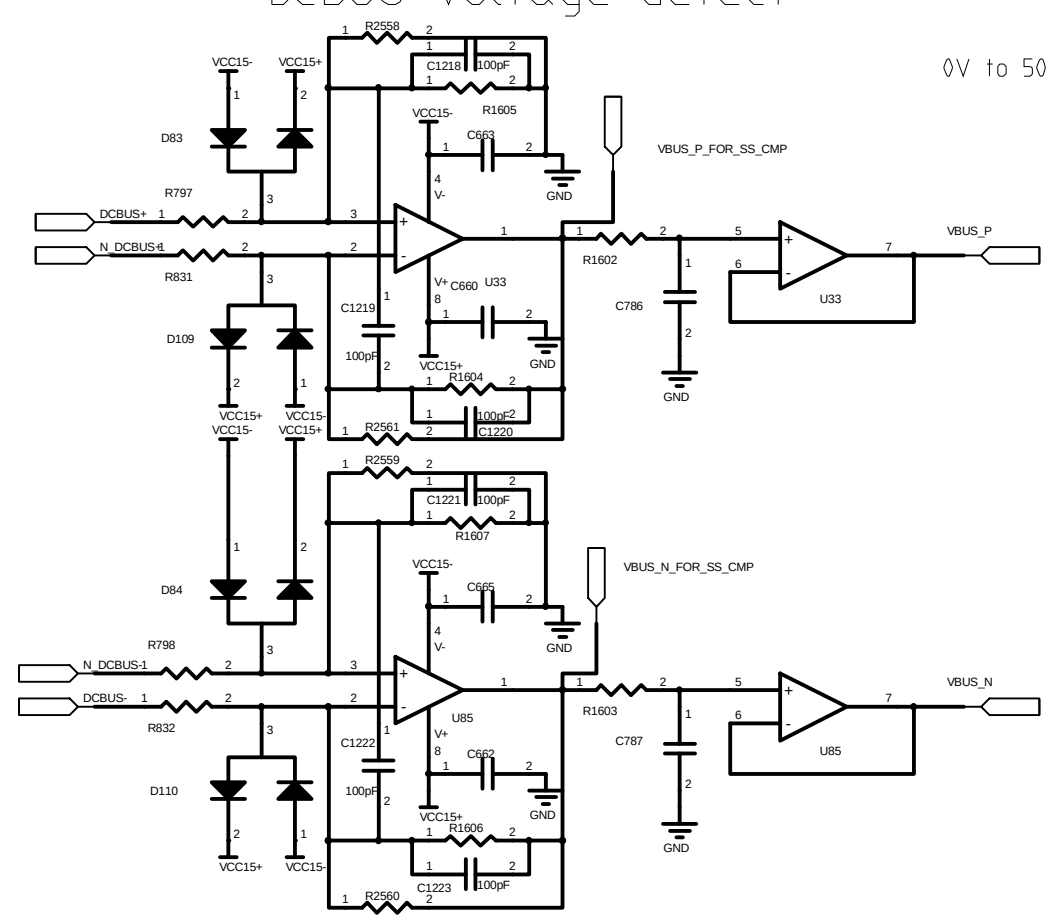
Total Output current Detect

0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO. xxx	SHEET. 13 / 36	TITLE. xxx Circuit Diagram	DRAWING NO. DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx			
Process Reviewed	xxx			
Standardized	xxx			
Approved	xxx			
		PRODUCT. xxx		
		PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		

1 2 3 4 5 6 7 8

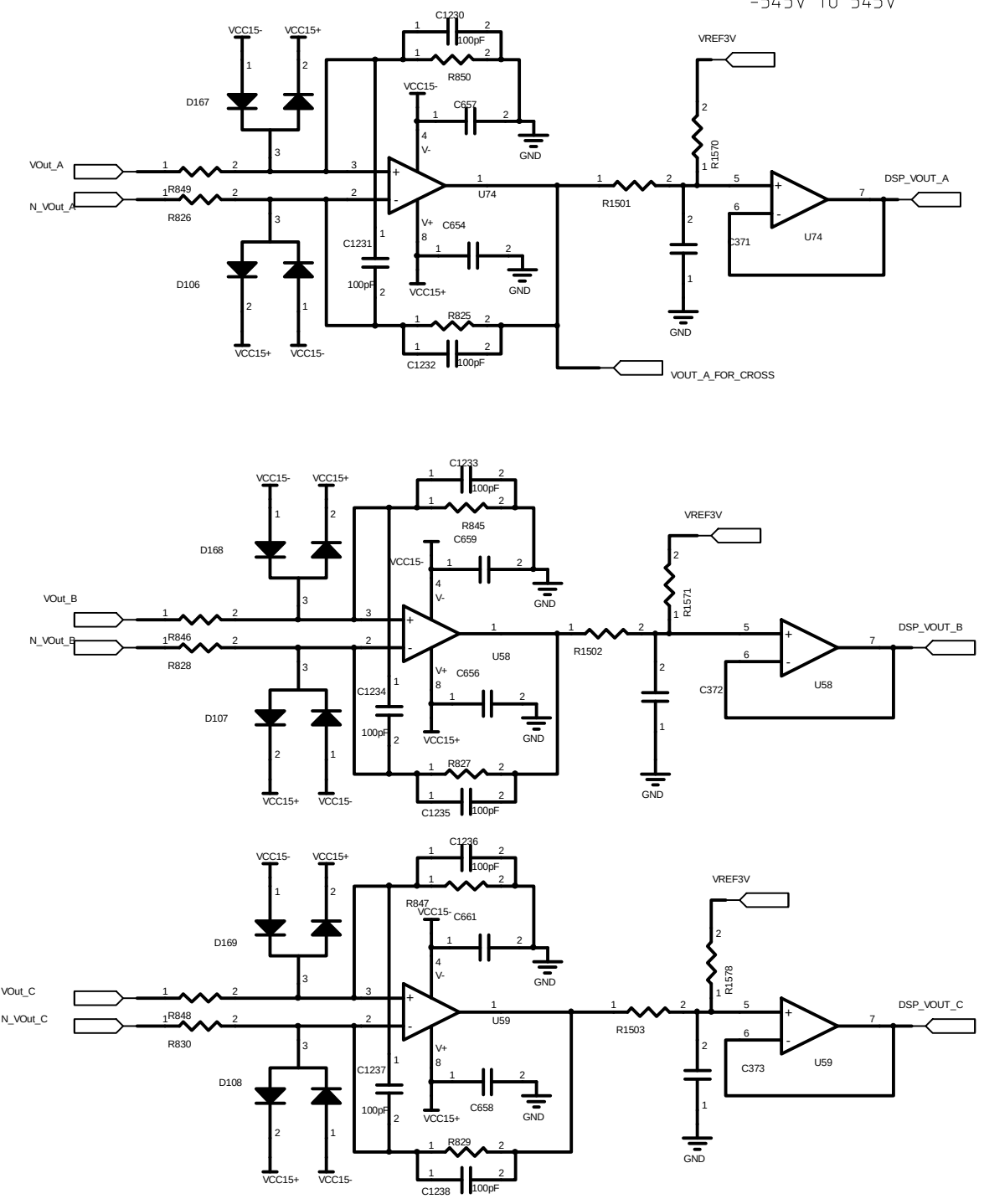
DCBUS voltage detect

0V to 507V



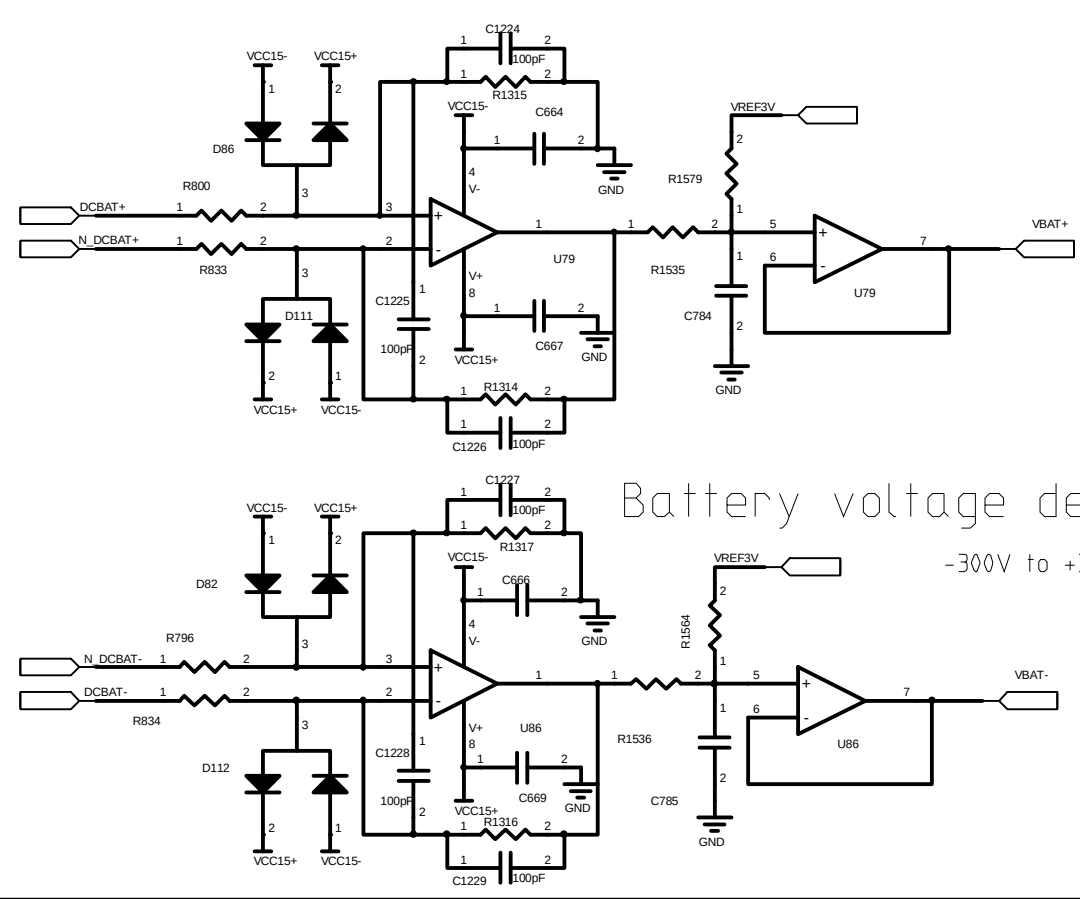
Output voltage detect

-545V to 545V



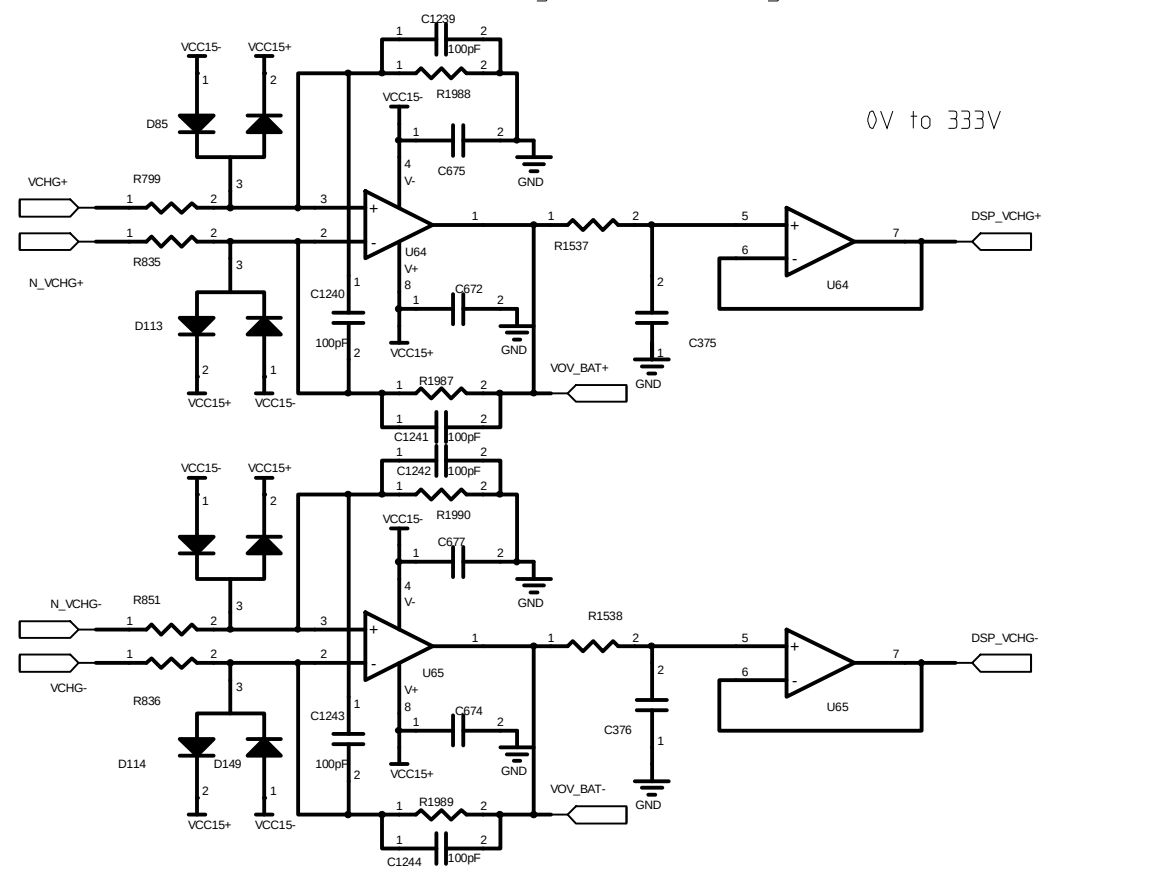
Battery voltage detect

-300V to +300V



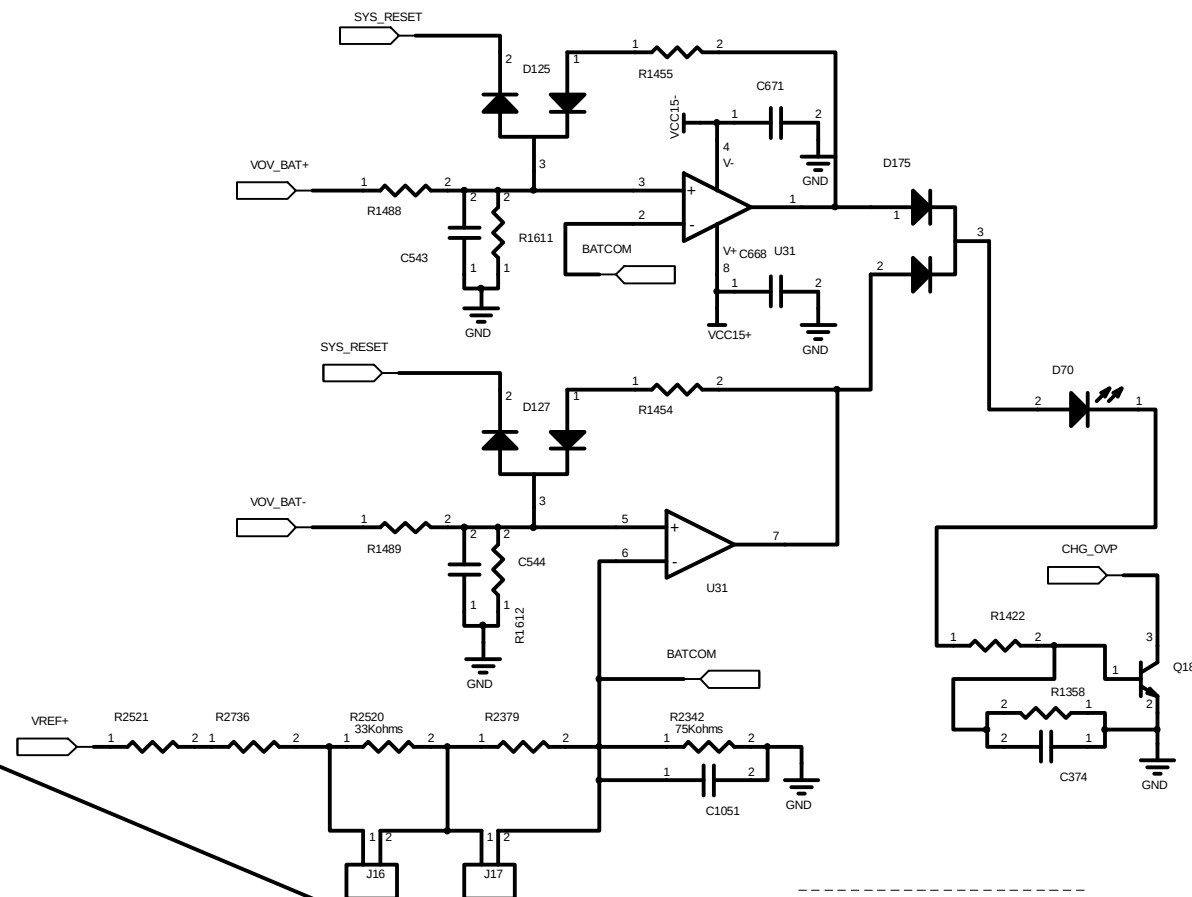
0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	14 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	XXX	
Process Reviewed	xxx		
Standardized	xxx		
Approved	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	

Charger voltage detect



Charge Overvoltage protect

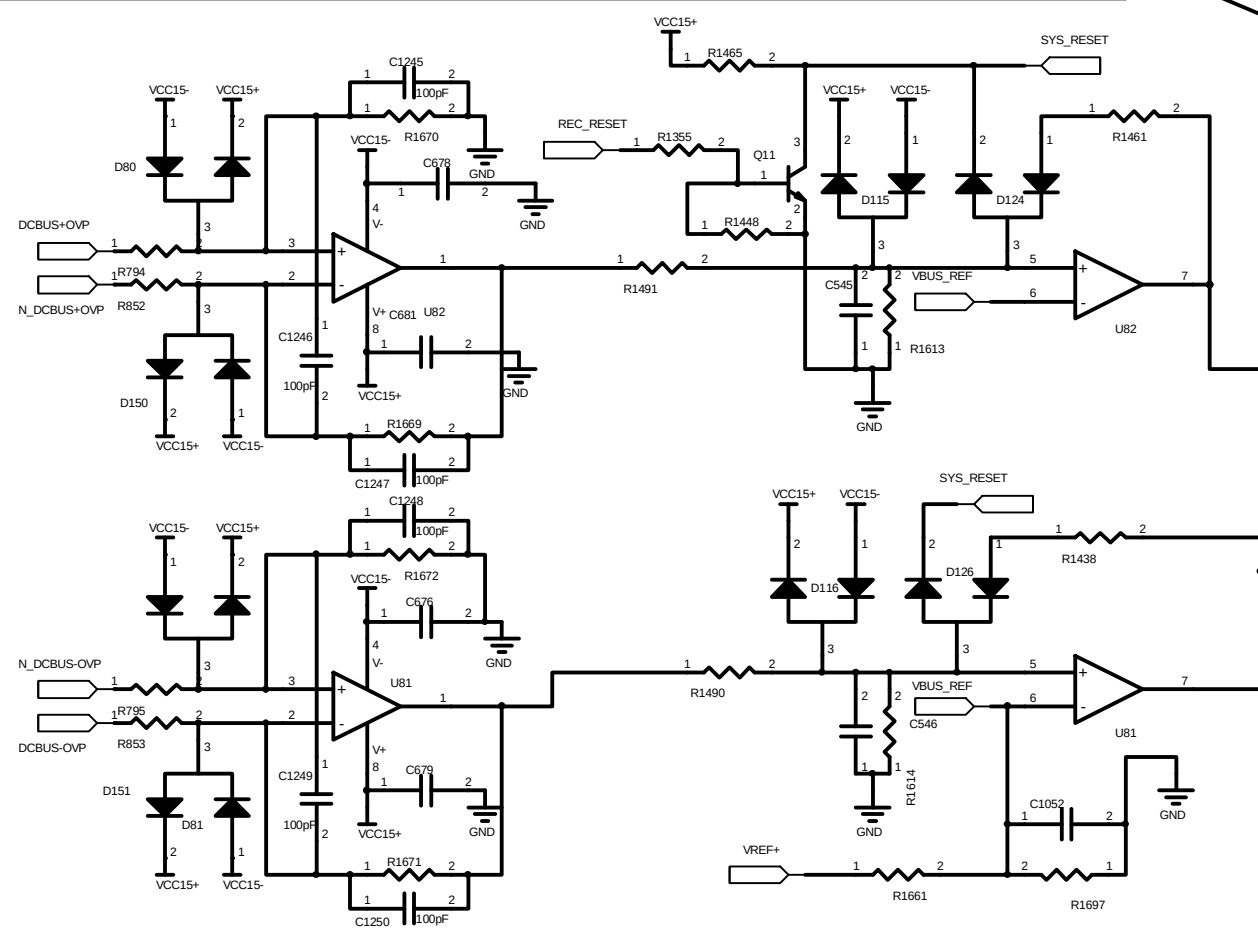
CMP Value : 236V



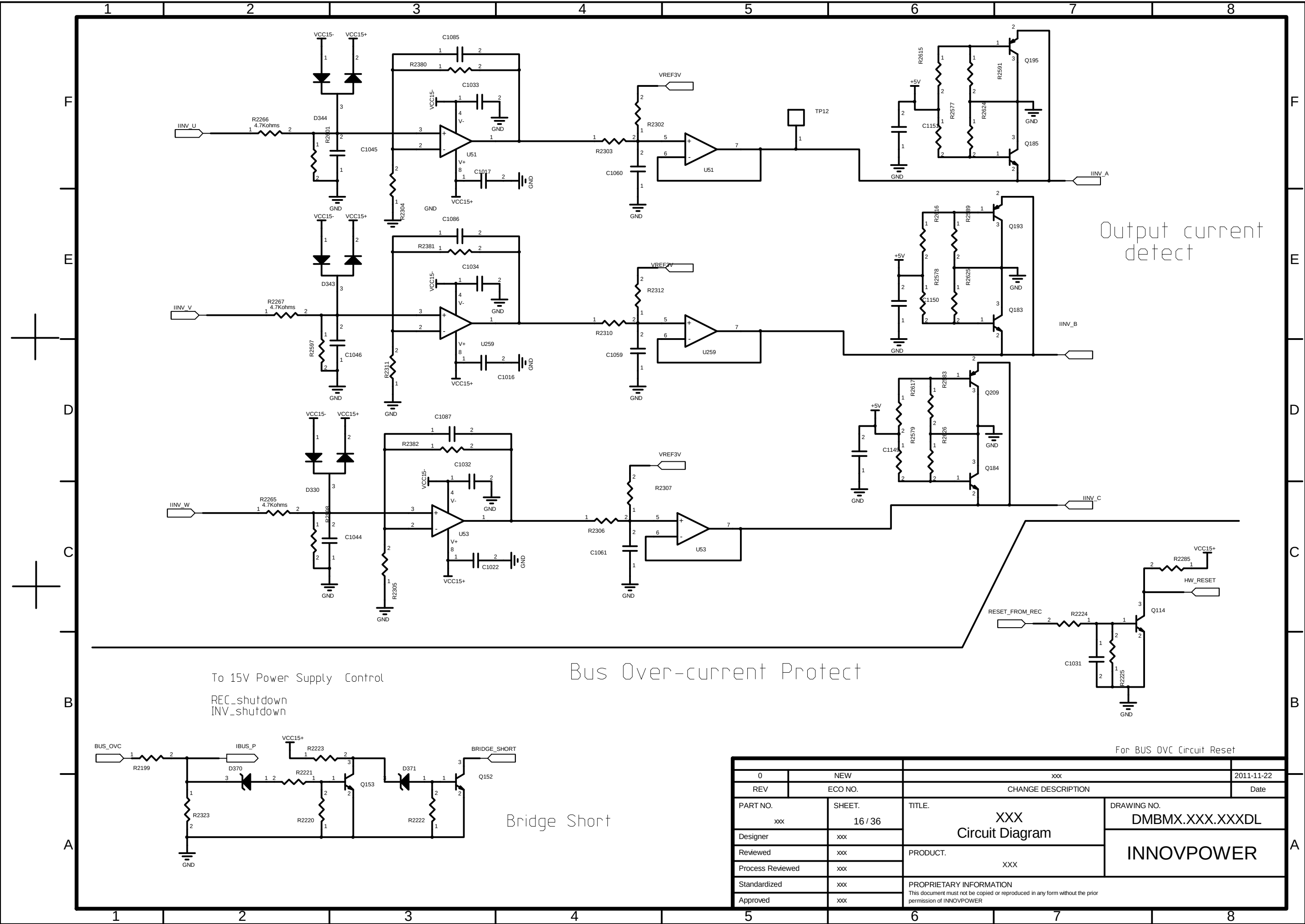
Bat nums: J16 J17
15 * * * *
16 * * ---
17 18 * *
19 20 ---

CMP Value : 484.5V

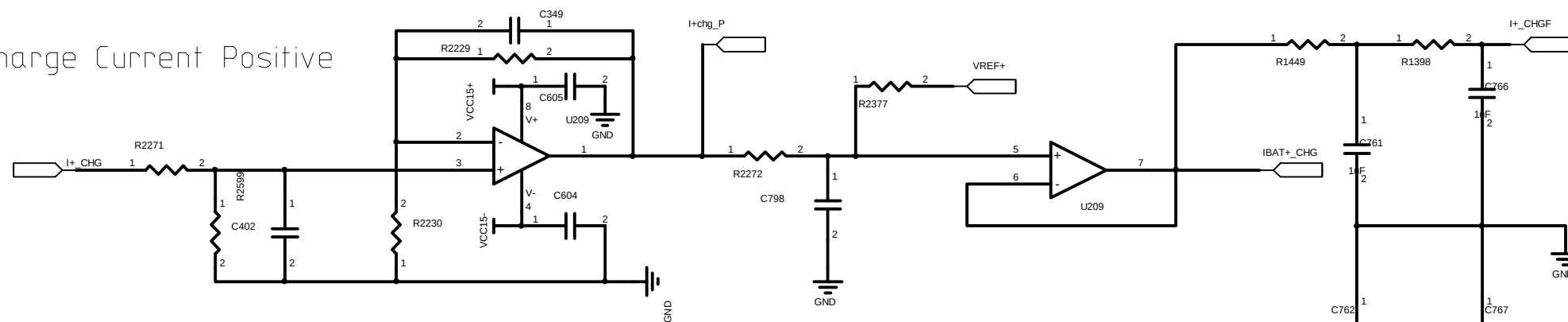
DCBUS Overvoltage protect



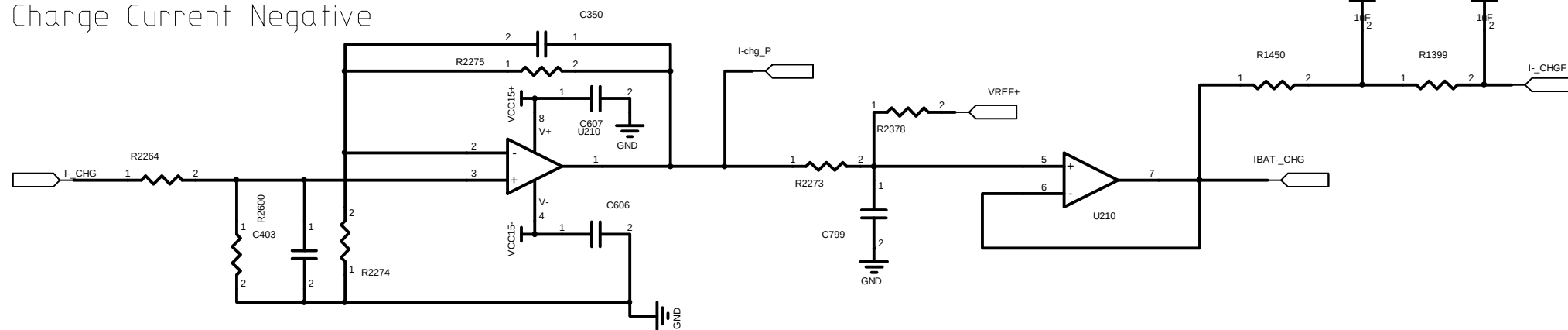
0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	15 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	XXX	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	



Battery Charge Current Positive

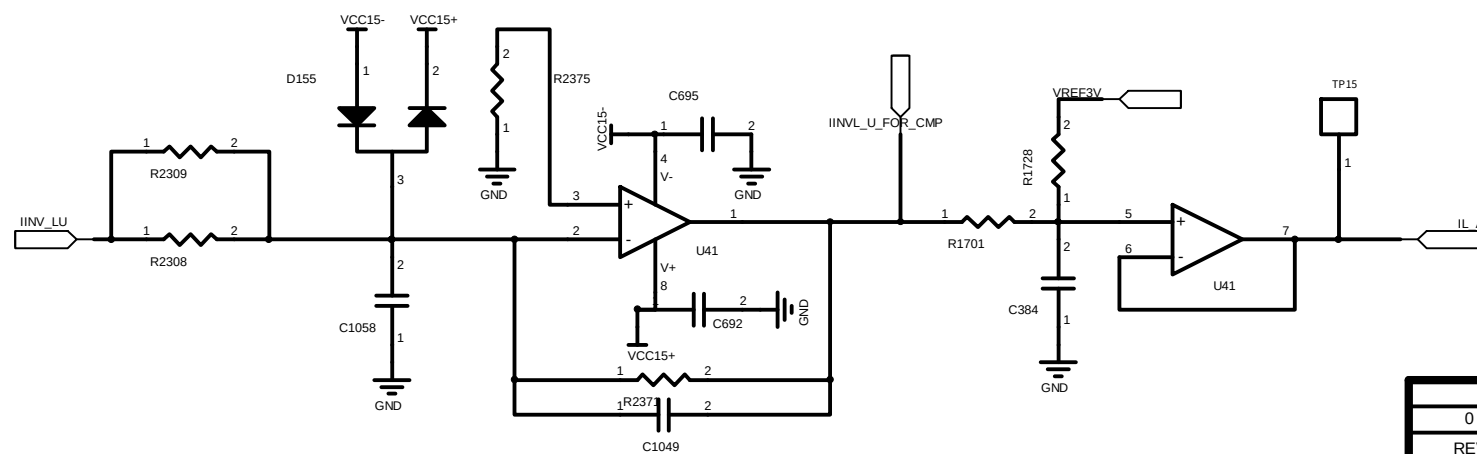


Battery Charge Current Negative

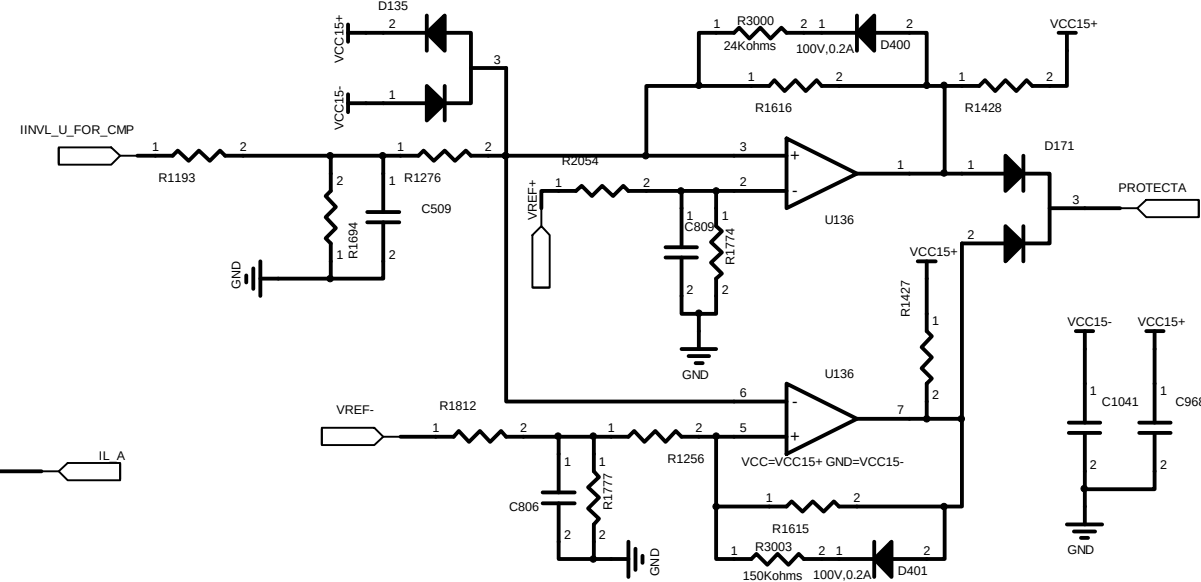


INV inductor current detect---

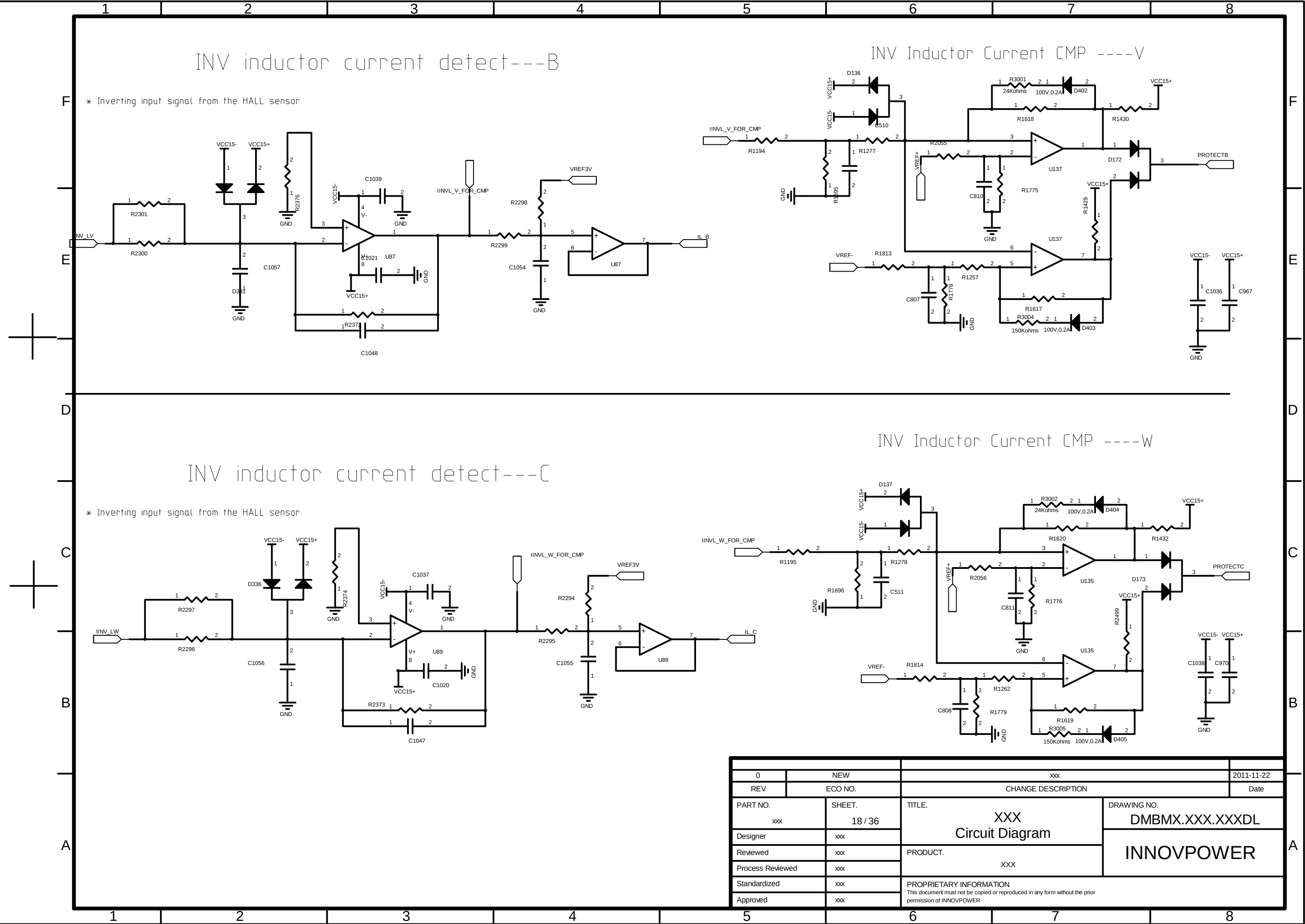
* Inverting input signal from the HALL sensor

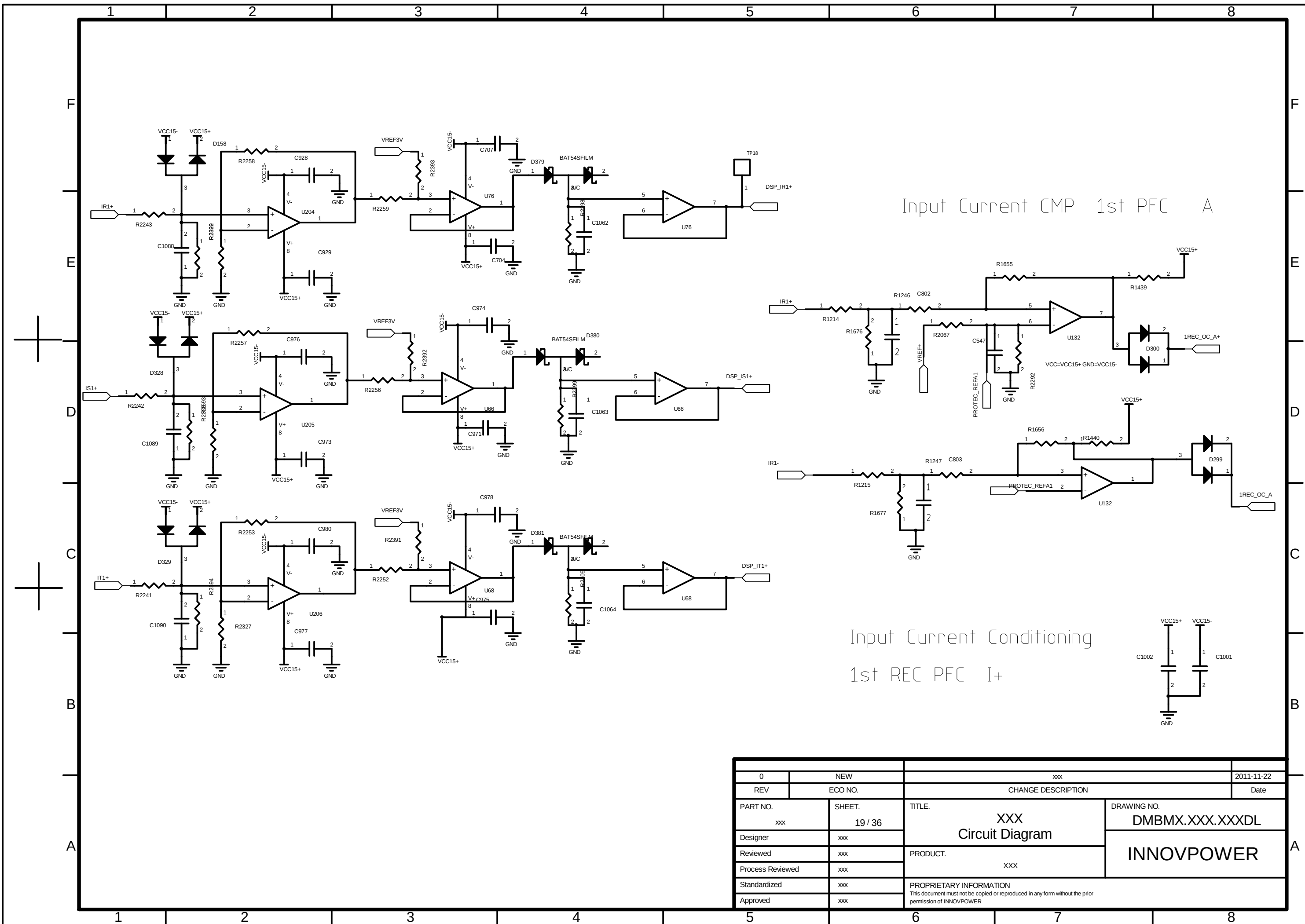


INV Inductor Current CMP ----U

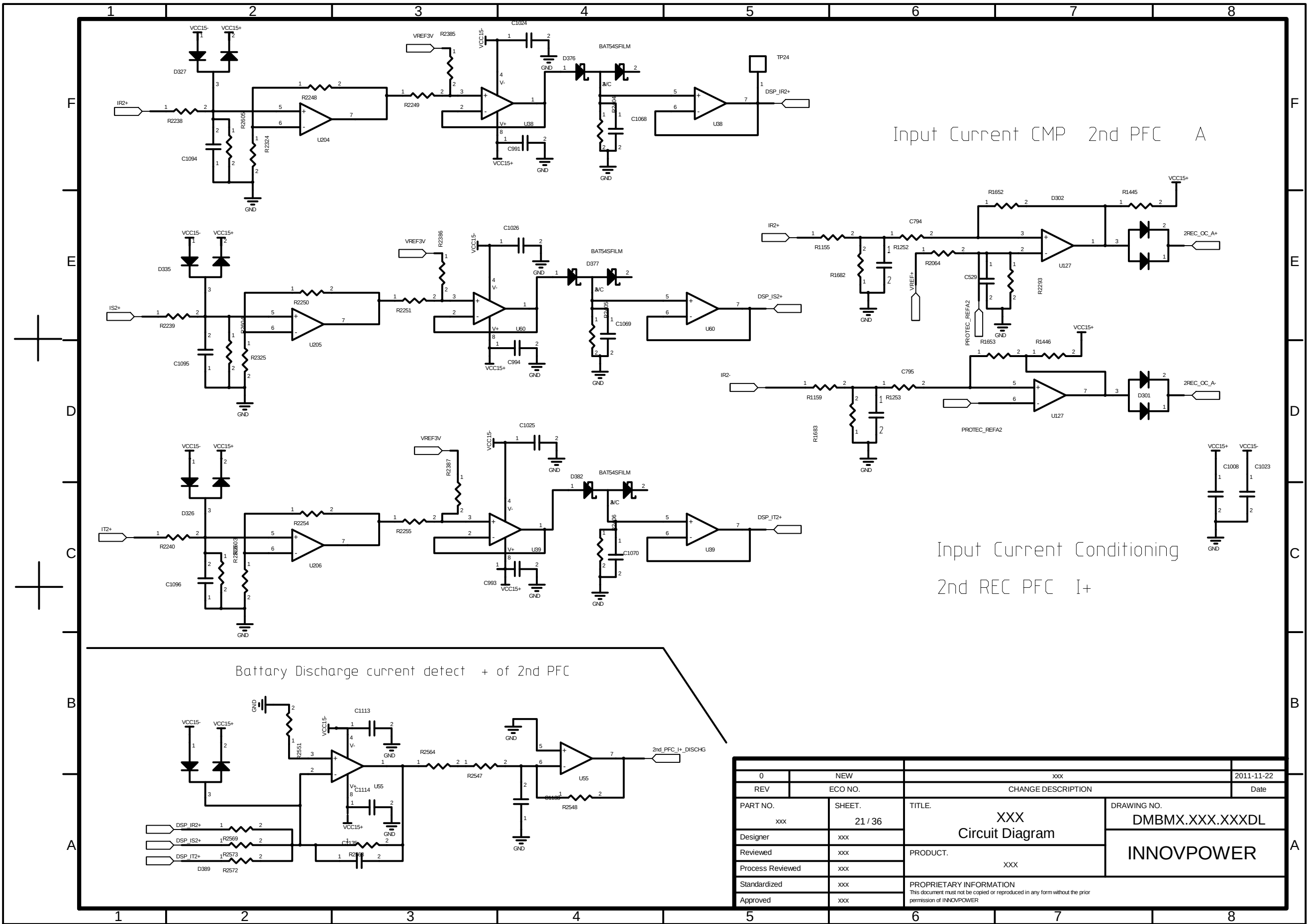


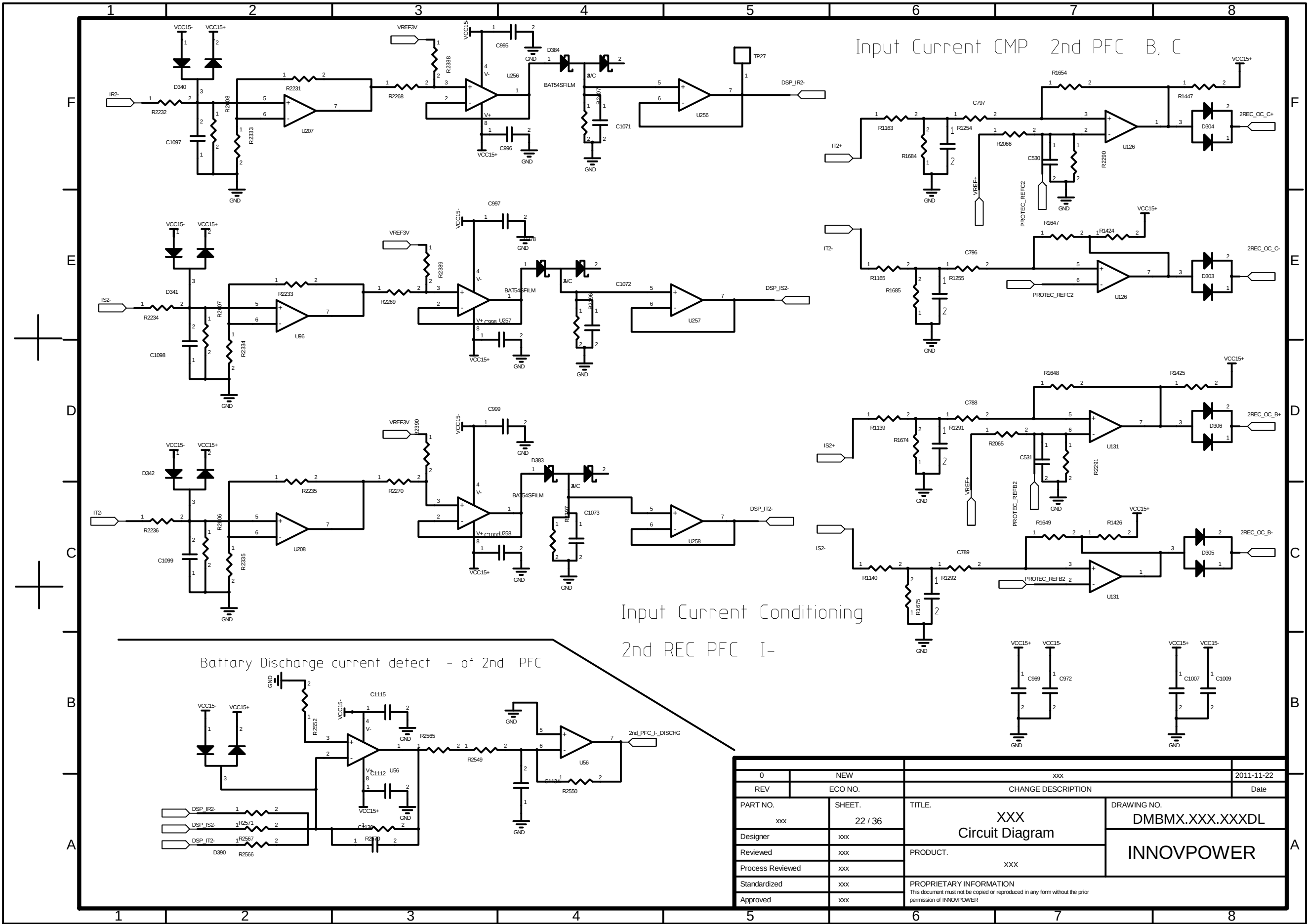
0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	17 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	

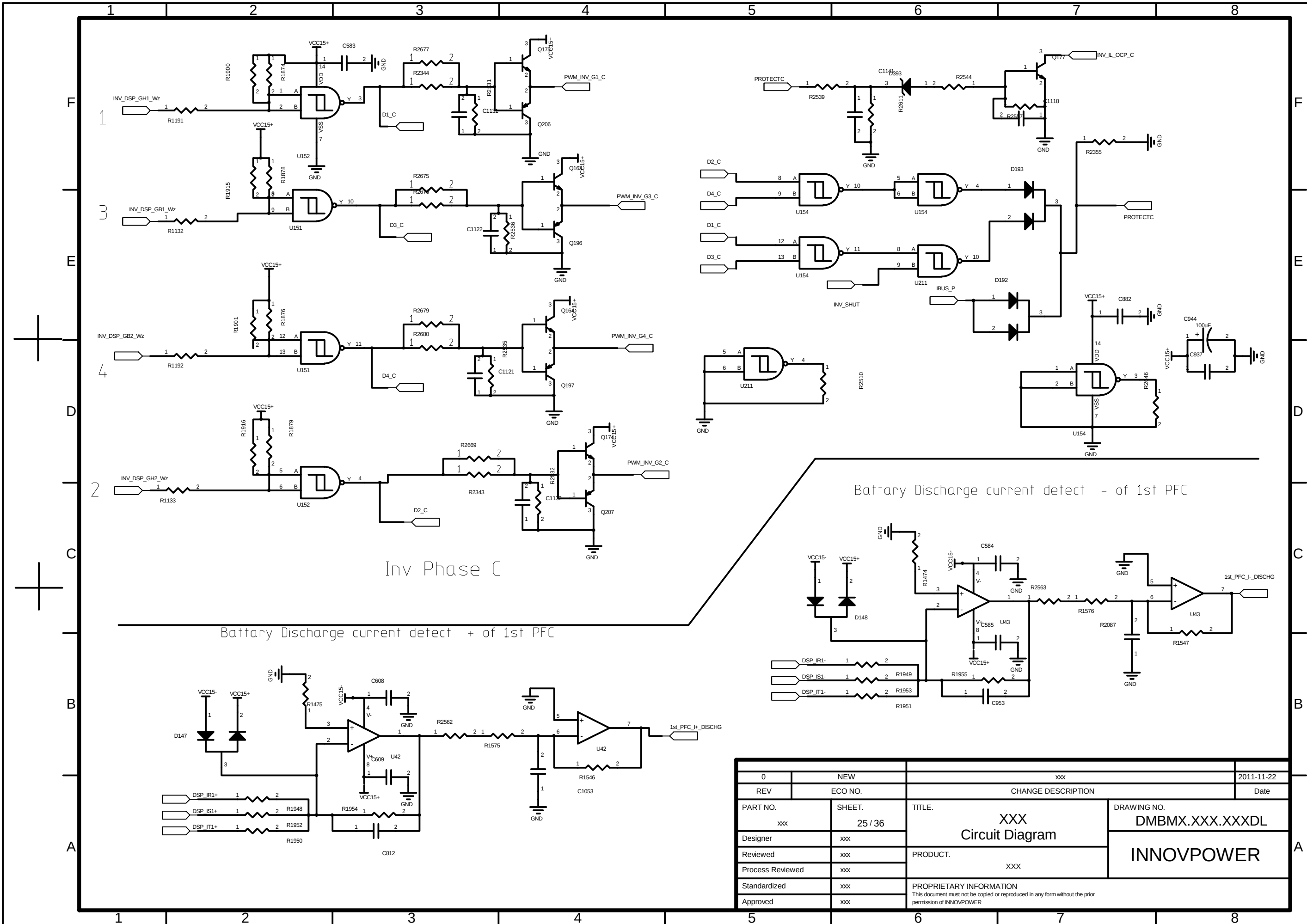




0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO. xxx	SHEET. 19 / 36	TITLE. XXX Circuit Diagram	DRAWING NO. DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx			
Process Reviewed	xxx			
Standardized	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		
Approved	xxx			





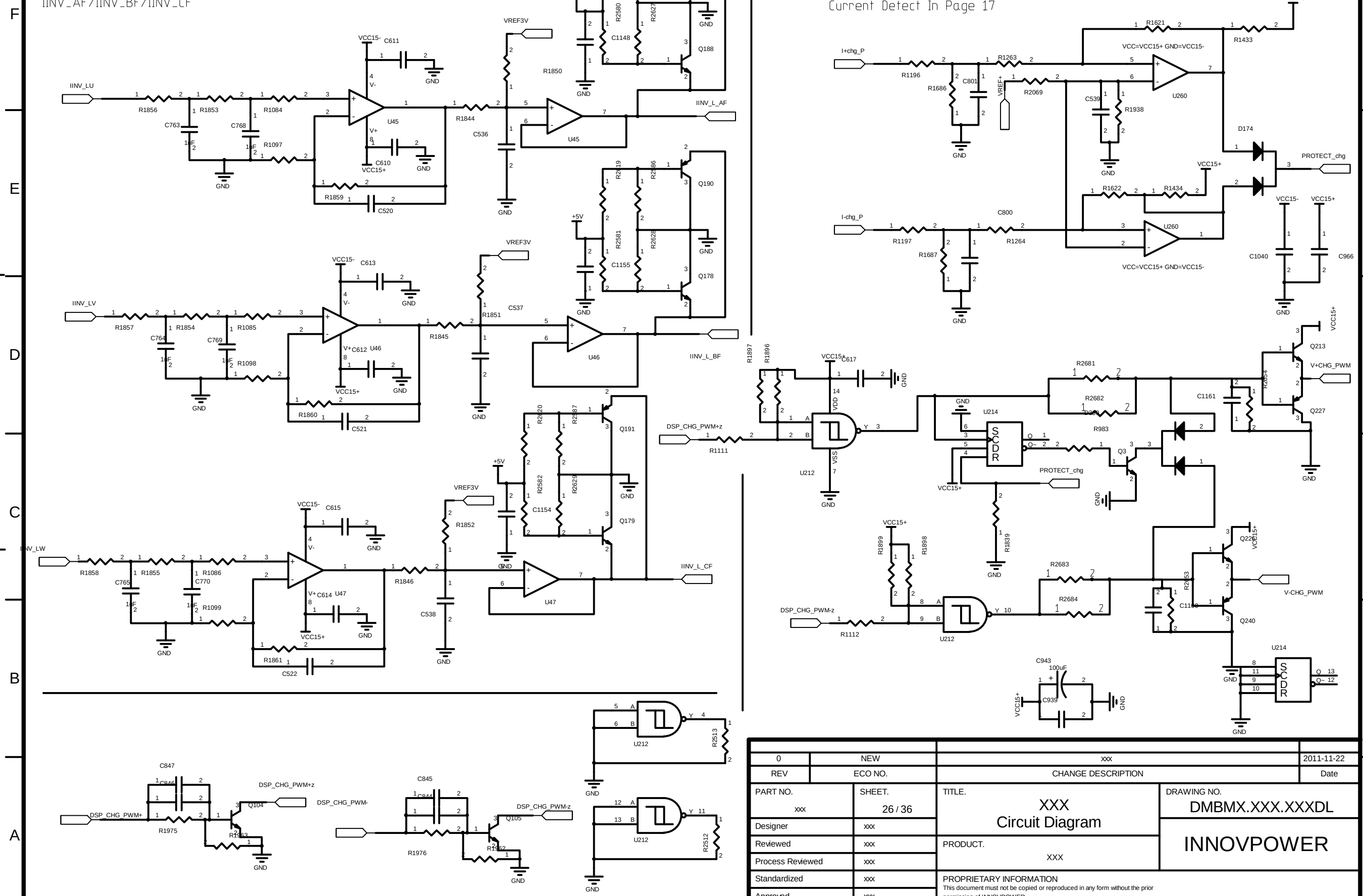


0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	25 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	

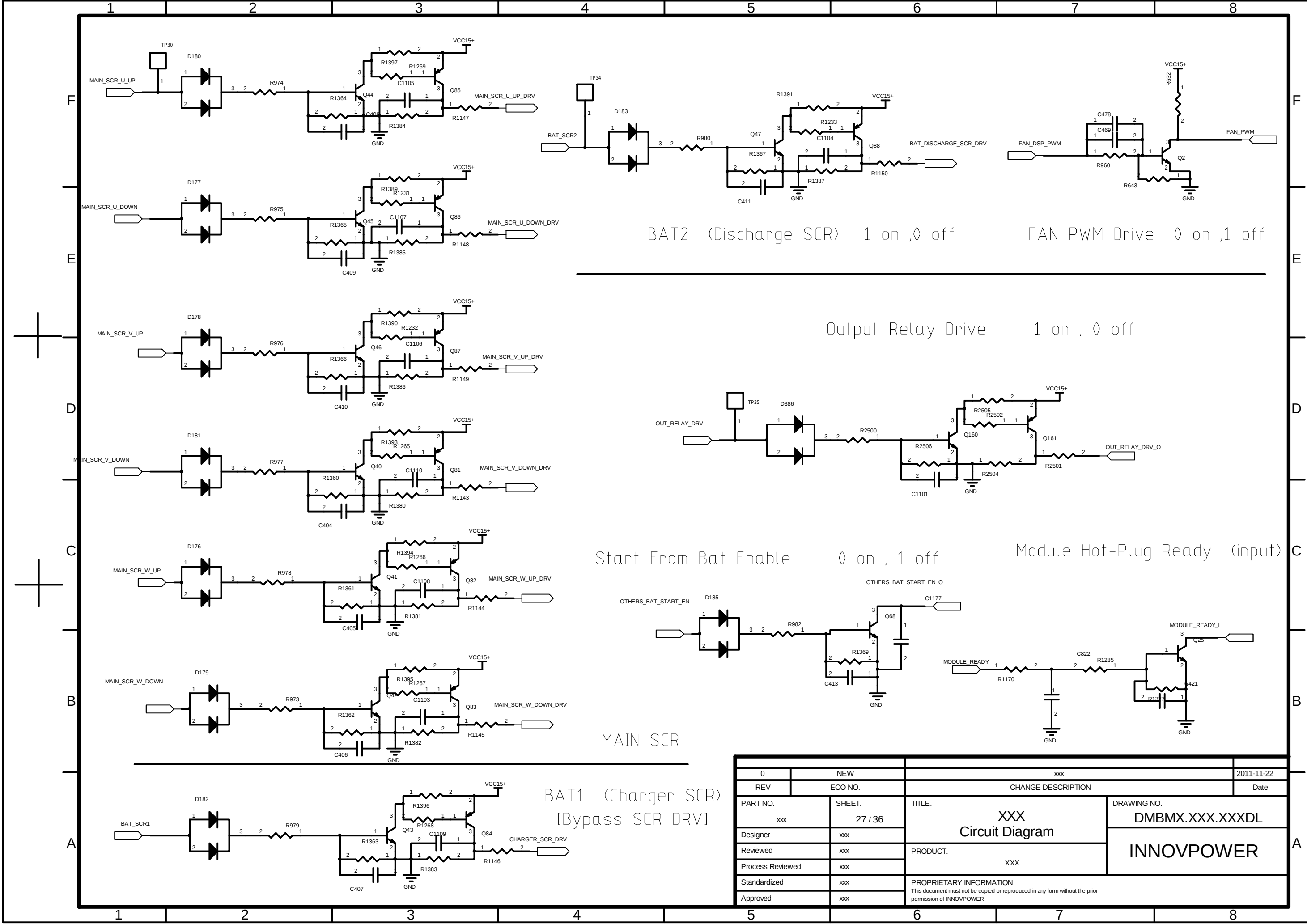
Output dc current Detect

IINV_AF/IINV_BF/IINV_CF

Charger Pulse by Pulse Limit
Limit Vaule : 25.3A
Current Detect In Page 17



0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	26 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	XXX	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	



BAT2 (Discharge SCR) 1 on ,0 off FAN PWM Drive 0 on ,1 off

Output Relay Drive 1 on , 0 off

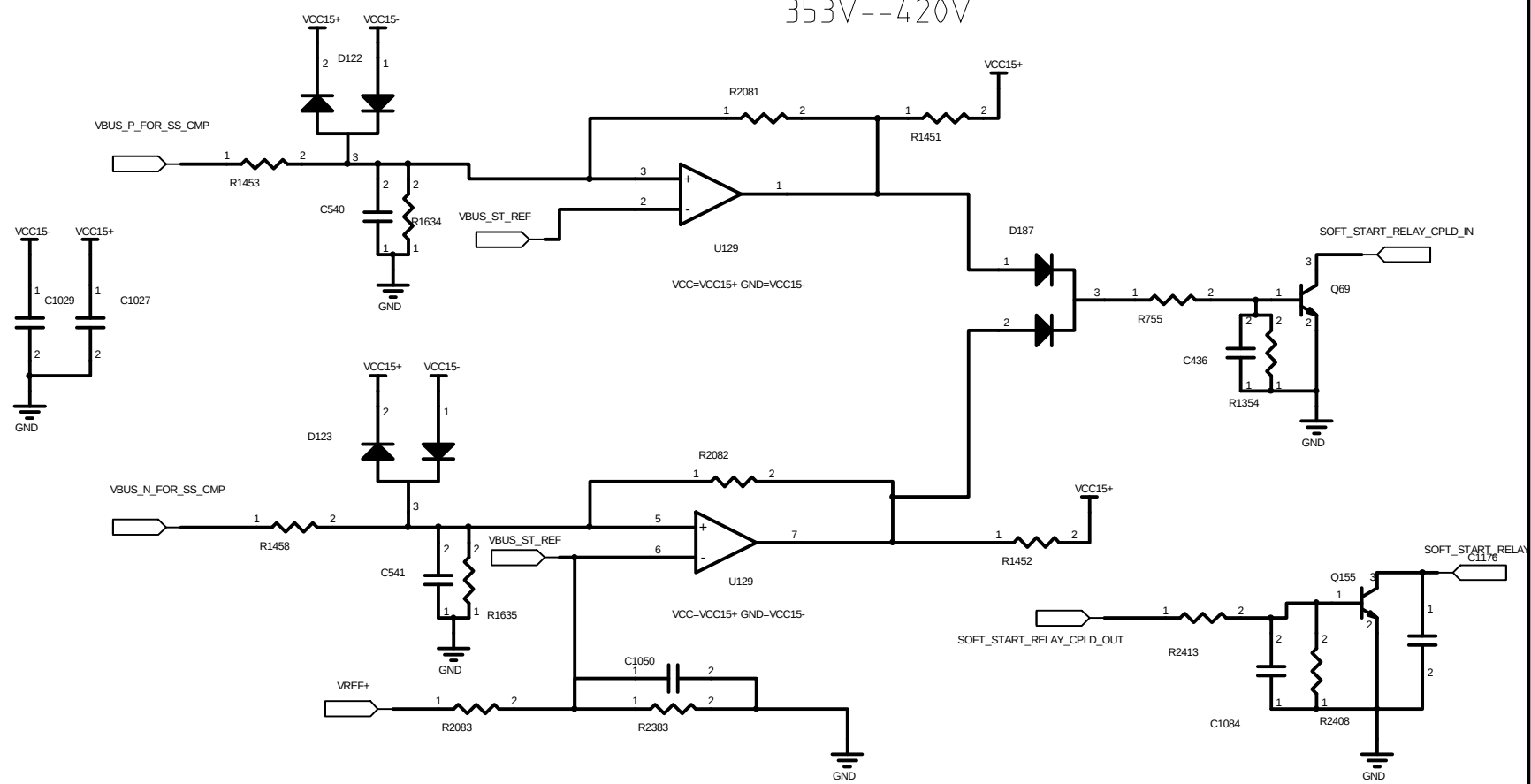
Start From Bat Enable 0 on , 1 off Module Hot-Plug Ready (input)

MAIN SCR

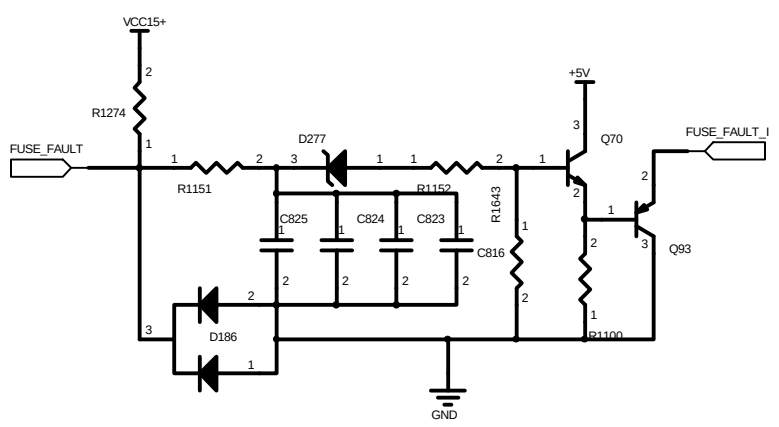
BAT1 (Charger SCR)
[Bypass SCR DRV]

0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO. xxx	SHEET. 27 / 36	TITLE. XXX Circuit Diagram	DRAWING NO. DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx			
Process Reviewed	xxx			
Standardized	xxx			
Approved	xxx	PRODUCT. XXX		
		PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		

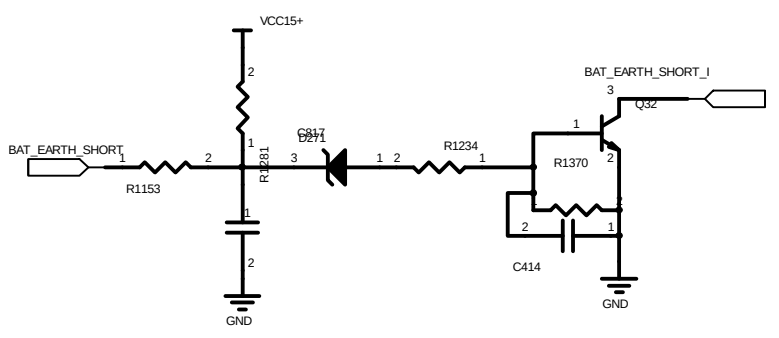
SoftStart Relay Drive (output)
353V--420V



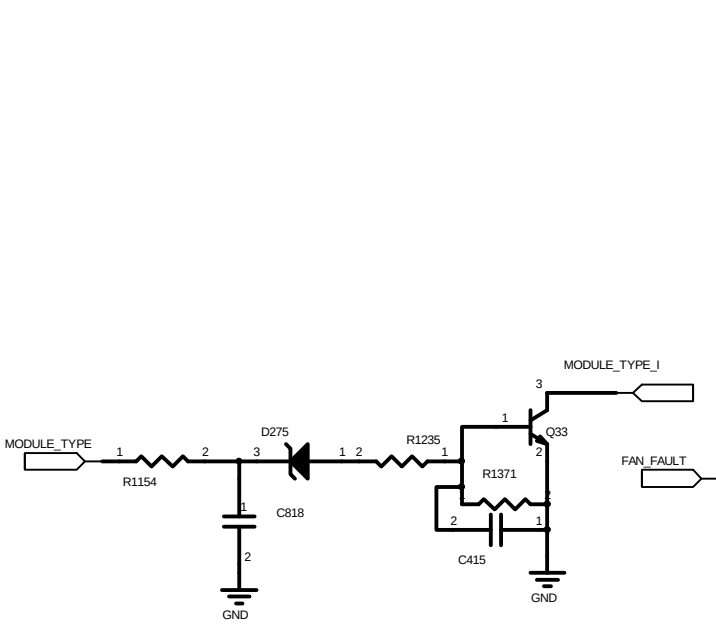
Fuse Fault (input)



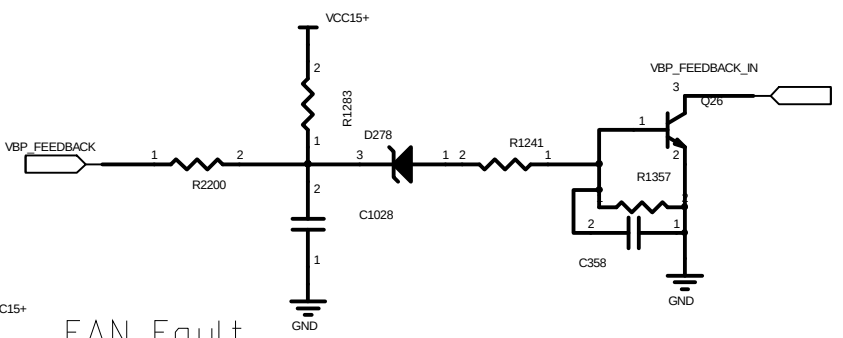
Bat-Earth Short Fault (input)



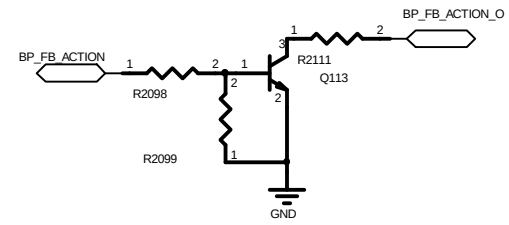
Power Fault (input) 5.1V---Fault, 0V---OK



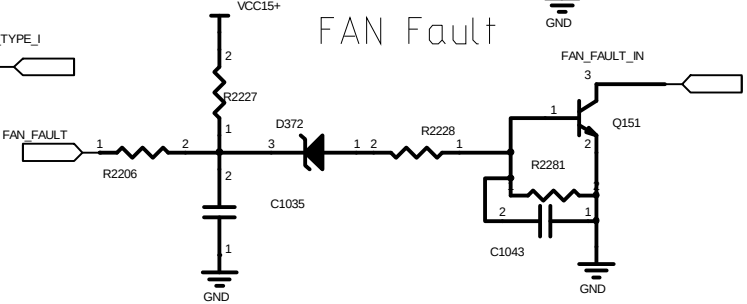
Bypass Feedback Reserved Input



Bypass Feedback Action Reserved



FAN Fault

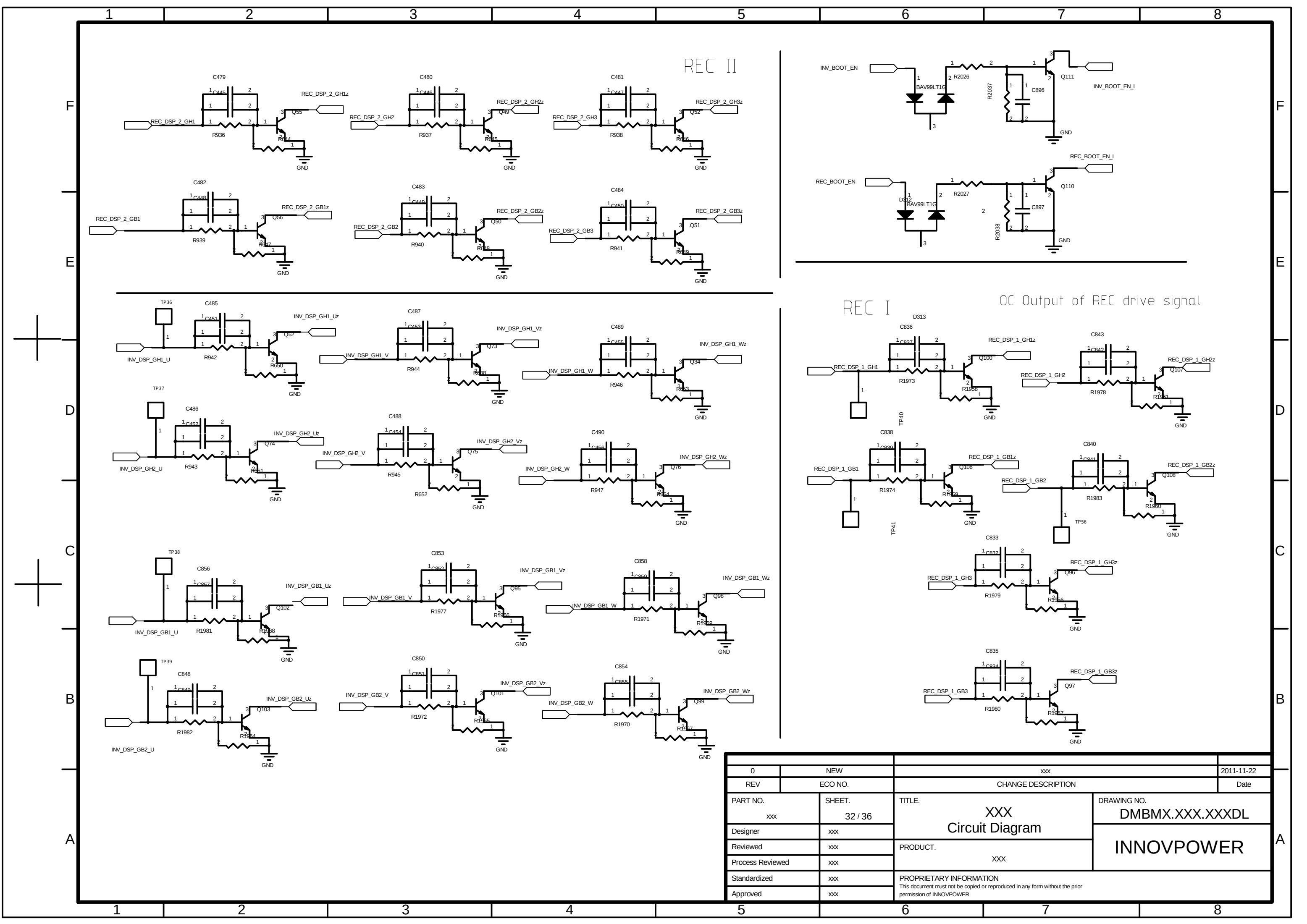


Module Type Select
(PowerModule Or BypassModule)

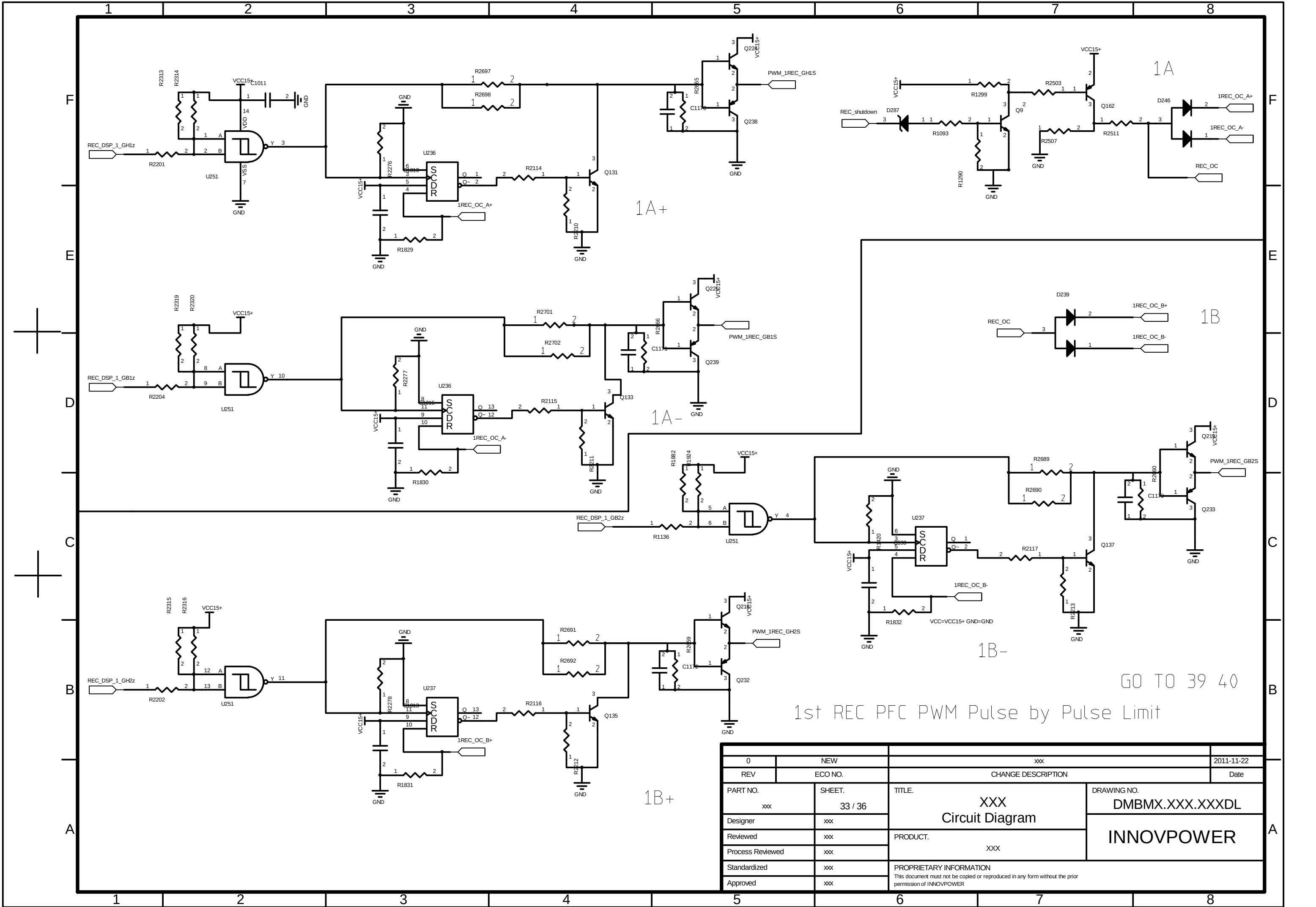
0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	28 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	



0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO. xxx		TITLE. XXX Circuit Diagram		DRAWING NO. DMBMX.XXX.XXXDL
Designer	SHEET. 30 / 36			INN OVPOWER
Reviewed	xxx	PRODUCT. xxx		
Process Reviewed	xxx			
Standardized	xxx			
Approved	xxx			
		PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		



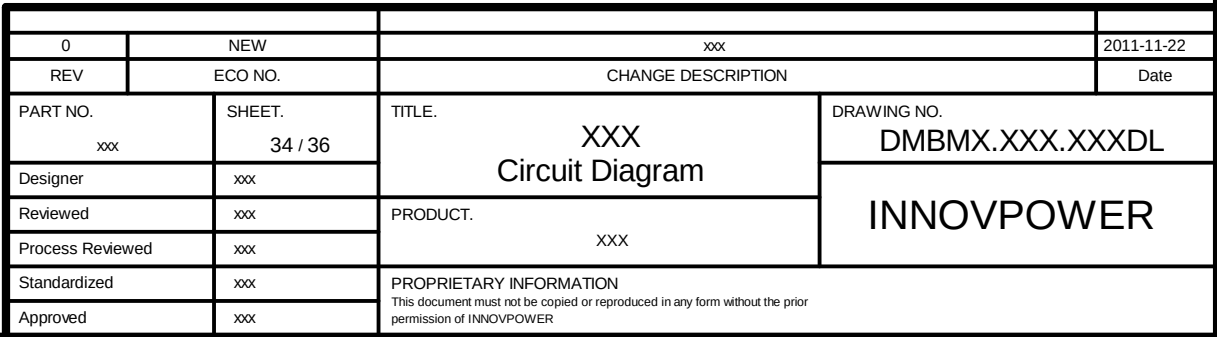
0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO.	SHEET.	TITLE. XXX Circuit Diagram	DRAWING NO.	
xxx	32 / 36		DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx			
Process Reviewed	xxx			
Standardized	xxx	PROPRIETARY INFORMATION		
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		

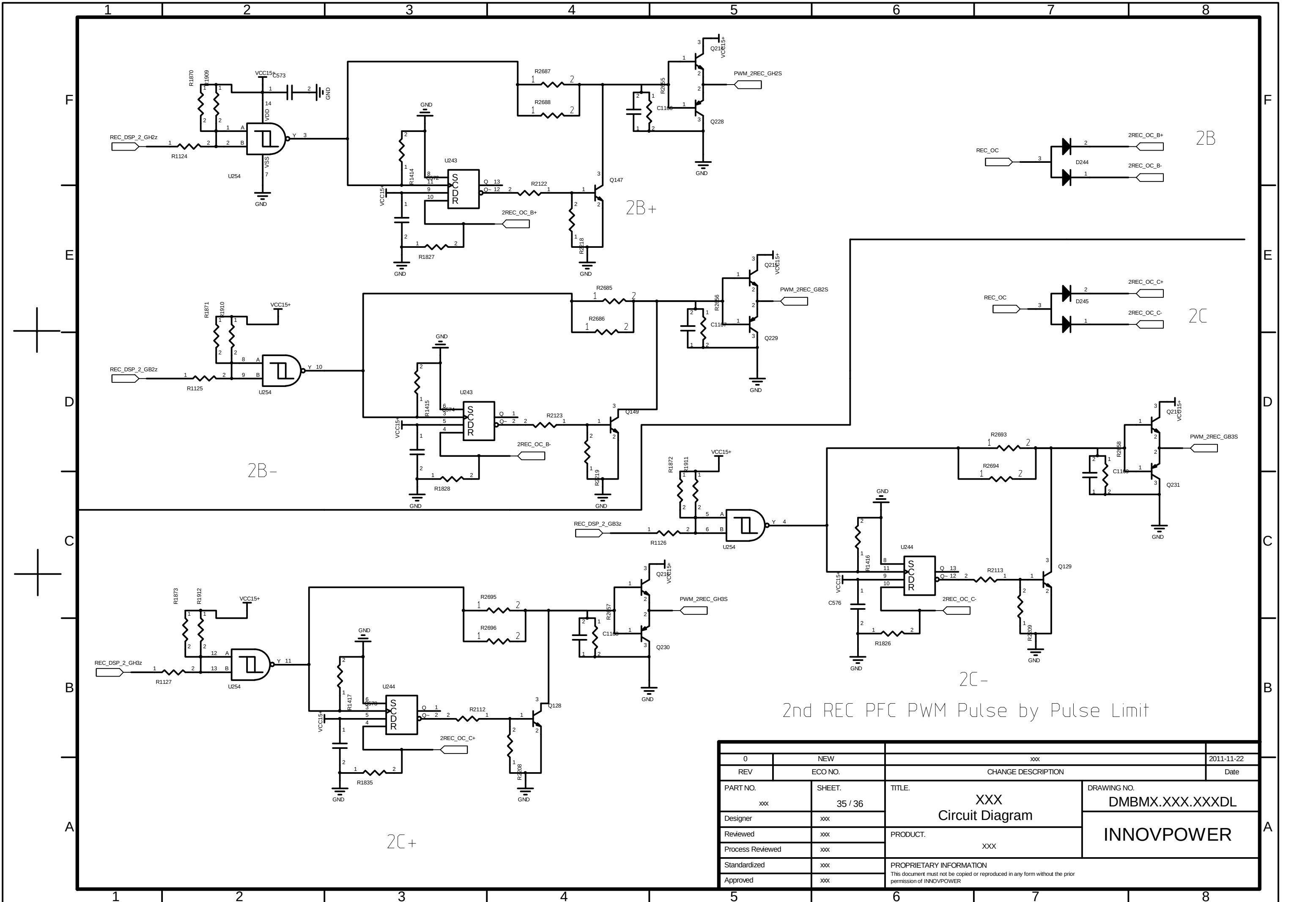


GO TO 39 40

1st REC PFC PWM Pulse by Pulse Limit

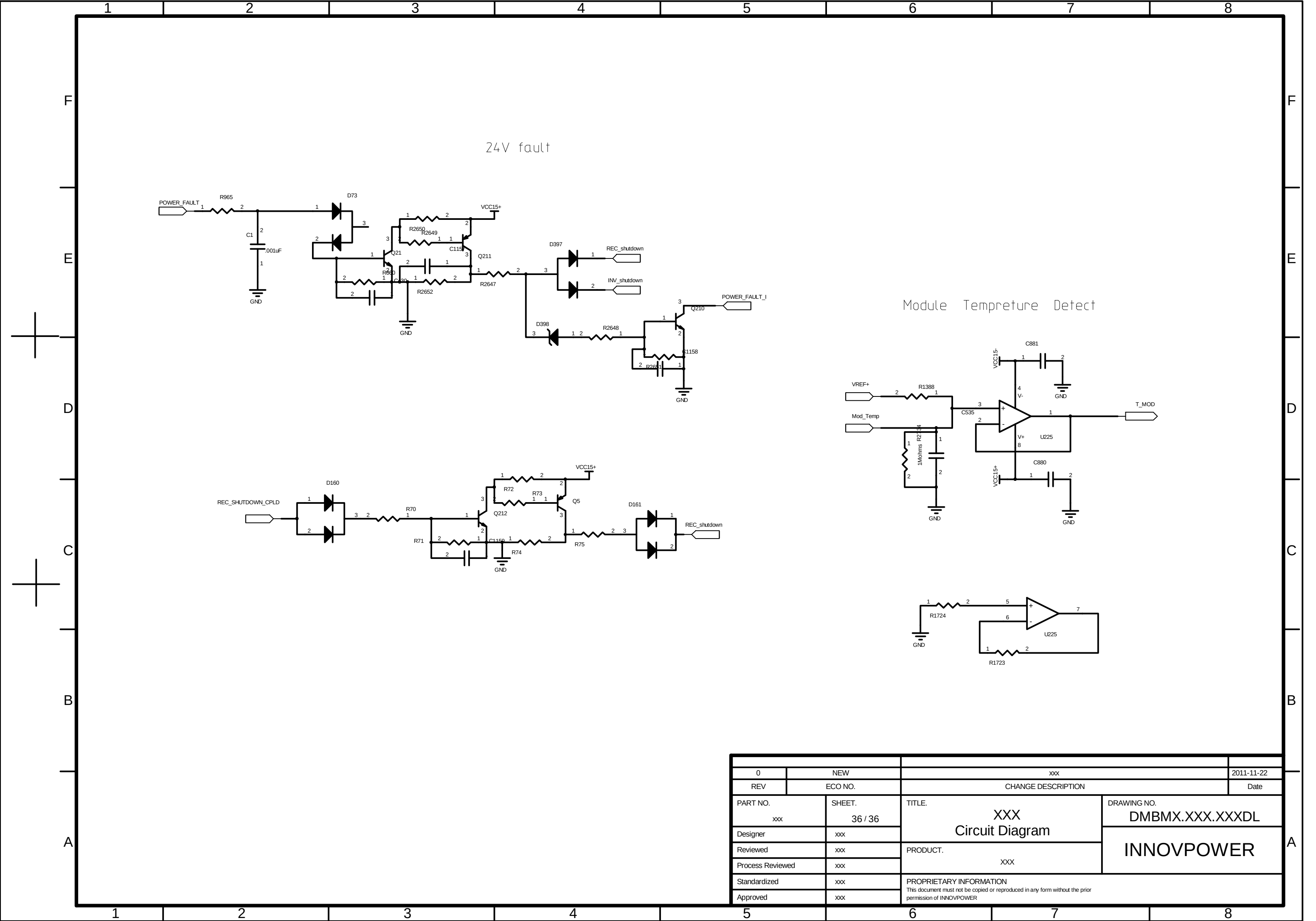
0	NEW	xxx	2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION	Date
PART NO.	SHEET.	TITLE.	DRAWING NO.
xxx	33 / 36	XXX Circuit Diagram	DMBMX.XXX.XXXDL
Designer	xxx	PRODUCT.	INNOVPOWER
Reviewed	xxx	xxx	
Process Reviewed	xxx		
Standardized	xxx	PROPRIETARY INFORMATION	
Approved	xxx	This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER	





2nd REC PFC PWM Pulse by Pulse Limit

0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO.	SHEET.	TITLE. XXX Circuit Diagram	DRAWING NO.	
xxx	35 / 36		DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx			
Process Reviewed	xxx			
Standardized	xxx	PRODUCT. xxx		
Approved	xxx			
		PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		



0	NEW	xxx		2011-11-22
REV	ECO NO.	CHANGE DESCRIPTION		Date
PART NO. xxx	SHEET. 36 / 36	TITLE. XXX Circuit Diagram	DRAWING NO. DMBMX.XXX.XXXDL	
Designer	xxx		INNOVPOWER	
Reviewed	xxx	PRODUCT. xxx		
Process Reviewed	xxx			
Standardized	xxx	PROPRIETARY INFORMATION This document must not be copied or reproduced in any form without the prior permission of INNOVPOWER		
Approved	xxx			